Probing the Limits of Memory: Can a Learned Response Persist Through Decapitation and Regeneration in Planaria?

By

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Abstract

Research stemming from the invertebrate literature is forcing us to question some of our assumptions about the nature of memory. Planaria, a flatworm with a centralised brain and incredible regenerative capabilities, have demonstrated retention of simple associative memories, such as associating a texture with a reward, in brainless tail fragments after decapitation. This suggests basic memories may be stored outside the brain but leaves open the question of whether complex goal-oriented memories share this capacity. To address this, we performed a series of experiments to determine whether planaria can acquire an operantly conditioned response that persists for at least two weeks, and whether this can be retained in the brainless tail halves of decapitated planaria. Using a Y-maze paradigm, we established baseline arm preferences and then rewarded treatment subjects with either cocaine or methamphetamine for entering their non-preferred arm during conditioning. Control subjects received vehicle only (distilled water). For the key experiments, subjects were then cut into head and tail fragments and allowed to regenerate for 14 days before we tested for memory retention. The next day, subjects were exposed to the rewarding compounds to identify whether the memory could be brought back or strengthened with a reinstatement procedure. Our results regarding whether planaria can learn and retain an operantly conditioned response were mixed. Experiments 3 and 4 provided preliminary evidence for learning, as treatment subjects entered the active arm more often at the end of conditioning. However, experiments 2 and 5 failed to show a significant change in behaviour compared to control subjects. We found no conclusive evidence that learned responses were retained in the brainless tail fragments, although the relatively weak learning may have limited the likelihood of successful retention. Future experiments will require more robust training methods to conclusively test the hypothesis of complex memory retention outside of the brain.

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Introduction

A brain in isolation is just a clump of extravagant cells. A brain earns its keep by liaising with the body and the external world. It is among these brain-environment interactions that an organism can set and achieve goals and, ultimately, carve a pathway to survival. But brains operate in the dark. Their only insight into the goings-on of the world is through delicately placed sensory organs such as the eyes, nose and ears.

The sensory technology that each organism possesses, which philosophers call its sensorium, differs across species. Some build a picture of the world using light-sensitive proteins. Others live where no light can penetrate and so must form their worldview using other sensory modalities like echolocation. Notwithstanding these differences, neuroscientists and biologists seek to understand the suite of abilities each organism possesses, the neuronal and molecular mechanisms which underpin these, and the factors that determine when and why an organism deploys the behaviours in its arsenal. But we do not usually do this for the sake of the organism itself. Rather, we use non-human organisms with the hope of learning something about our own brains and bodies.

We now have a broad set of tools for inspecting brains across different time spans and at different levels of analysis. From activity within a single dendritic spine over microseconds to connectivity between different brain structures over several minutes. We can even track changes in the size of spines on a single dendrite over time – impressive given the width of a spine is 100 times smaller than the thickness of a human hair (Bayramoglu et al., 2022; B.-Z. Li et al., 2023). At the network level, we are able to identify groups of neurons (ensembles) involved in encoding and storing a memory, and can use precise tools to excite or inhibit those networks to alter an animal's behaviour (Goshen, 2014; Liu et al., 2012).

Our experimental competency arose from many small steps. Before we had the capability for manipulating neurons to understand their role in memory, we had to attack things more abstractly. Our early endeavours to understand memory involved basic procedures like learning lists of nonsense syllables or simple motor tasks (Ebbinghaus, 1913; Gabrieli et al., 1993). Very basic stuff. But this early research helped answer the question of whether memory is a unitary system or a suite of separate systems which can be dissociated. Out of this fell the distinction between episodic and semantic memory, as well as short- and long-term memory storage. Early theoretical progress provided the foundation upon which specialised tools and procedures could be developed. With these, we can now investigate and characterise the biology of memory in its different forms.

Overview of Key Concepts in the Field of Learning and Memory

Categories of Memory

Memory is the embodiment of past experience which shapes future behaviour. Learning, on the other hand, is the process of memory acquisition. That said, there may be as many different definitions of learning and memory as there are papers published on the topic. Barron et al. (2015) surveyed the various uses of the term "learning" across disciplines including cognitive psychology and behavioural ecology and identified at least 50 definitions (albeit with a lot of overlap). Despite this, there is general agreement that memory can be parcelled into several distinct categories based on the content of the information being held (reviewed by Camina & Guell, 2017). As can be seen in Figure 1, a major distinction is made between explicit and implicit memory (Schacter & Tulving, 1994; Squire, 1987). Explicit memories are those accessible to conscious awareness, like the memory of where you parked your car this morning. Implicit memories cannot be consciously accessed but still affect behaviour, an example being the muscle memory used to ride a bike. Explicit memory has been further subdivided into episodic and semantic memory (Tulving, 1972). Episodic refers to the rich experiential quality of personal memories, while semantic relates to things that you know but which lack an experiential component, such as facts about the world.

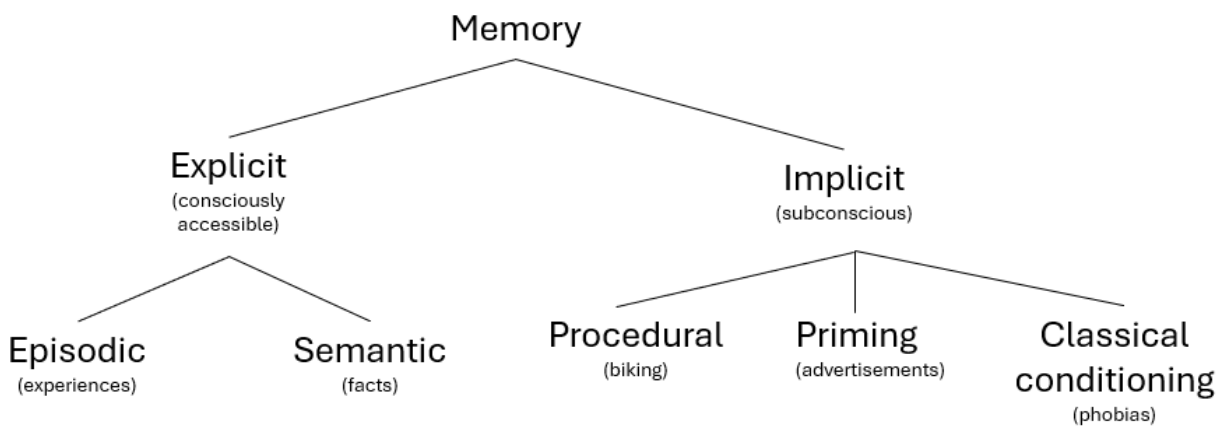
Memory can also be categorised temporally. Atkinson and Shiffrin (1968) proposed a model with three stores that process memories over time: a sensory register, short-term store, and long-term store. This division is still usefully applied in the field of learning and memory (e.g., J. A. Miller & Constantinidis, 2024). The temporal framing of memory reflects the process of learning itself. Learning is normally broken up into several stages, each building on prior stages to fortify the memory and increase its longevity. Initially, information we process enters a short-term state and may alter behaviour and decision making in the immediate future. However, most sensory information is not retained. Only meaningful information is given permanent residence. For this to occur, a set of active processes are required to ensure the information is maintained so that it can be accessed in perpetuity. This stage, known as consolidation, pushes back against the otherwise imminent process of forgetting.

These distinctions seem intuitive in the context of human memory. But it is not clear whether these distinctions generalise to other organisms. Most people do not attribute rich experiential memories to rodents, let alone invertebrates. Yet even simple organisms such as planaria have the capacity to retain and act on information from their environment (Deochand et al., 2018). Viewing the memory system as an array of dissociable information stores is particularly relevant when investigating whether information can be stored outside of the central nervous system. If, as recent research suggests, memories are able to be stored outside the brain in simple organisms (discussed later), the categories outlined here will help us to explore which forms of information have this property and which do not.

It may be that non-associative memories such as sensitisation and habituation can be stored outside of the brain, but perhaps rich episodic experiences can only be stored among finely tuned connections between neurons. We do not yet know the bounds of memory storage outside of the central nervous system. Armed with these conceptual distinctions of memory, researchers can investigate a broad range of training techniques among model organisms to identify which forms of memory (if any) persist outside the brain and which do not.

Figure 1

Categorisation of Memory



Note. Theoretical categorisation of memory based on the content and conscious accessibility of the information. The major explicit/implicit distinction was first put forward by Endel Tulving (1972). The examples provided for implicit forms of memory are intended to be illustrative, not exhaustive. A similar figure was presented in Squire (1987).

Associative and Non-associative Learning

As stated earlier, learning is the process of memory acquisition. But the type of information being acquired can differ, and so a variety of labels are used in the literature to capture this variability. Associative learning requires learning the temporal relationship between two stimuli. For example, learning that one stimulus reliably precedes another, or a behaviour reliably elicits a reward. Non-associative forms of learning capture learning about a stimulus itself, rather than its relationship to other stimuli. This typically takes the form of behavioural sensitisation or habituation. If you were to deliver a mild shock to Brett's hand, he would withdraw it reflexively as part of the innate startle response. But with repeated administration of the shock over time, Brett would learn that the shock is not harmful. The size of his startle response would decrease (habituation). He would have learned something about the shock, but learned nothing about its temporal relationship with other stimuli. To translate this example to an associative form of learning, a moderate shock could be delivered after Brett is shown a picture of a sunflower. He would learn an association between the flower image and the subsequent painful experience. With repeated pairings, Brett would display a pre-emptive startle response (tense his muscles, squint his eyes, dip his head) to presentations of the flower alone. He has learned a temporal association between the flower and the shock, such that his body now predicts and prepares for the shock before it arrives.

Classical conditioning and operant conditioning are two frequently used forms of associative learning. Classical conditioning involves learning an association between two or more stimuli, as in the flower/shock example above. Operant conditioning differs from classical conditioning in that rather than one stimulus being paired with another stimulus, a behaviour comes to be associated with a specific outcome. For example, Brett learned that signing up for psychology experiments often involves exposure to irritating stimuli. This changes the likelihood of that response being produced in the future. In other words, he stops signing up as a participant.

Maladaptive Learning

The examples outlined above capture learning in cases where it is beneficial for the learner. In the classical conditioning case described above, preparing for a shock by tensing his muscle tissue reduces the painfulness of the experience for Brett and minimises the chance of tissue damage. For the operant conditioning case, avoiding future experiments would mean he is exposed to less unnecessary irritants. Although these are mundane examples, we will all encounter consequential cases whereby our wellbeing and longevity are enhanced by learning from past experience. A close call when crossing a road (and the negative physiological experience that ensues) increases the likelihood of diligently looking both ways before future crossings.

The survival benefits of learning are obvious. Yet, the capacity for learning leaves us vulnerable to developing maladaptive habits. Consider Rebecca’s first experience with heroin. Prior to consumption, Rebecca has heard about heroin, but only in the sense that she knows it is a harmful drug, that similar drugs are used in a medical setting, and so on. She has no prior subjective experience of its effects. After several experiences with the drug, she learns about the intense sense of euphoria that comes from its consumption. Later, Rebecca develops a strong motivation to take the drug again, especially when she sees drug-associated cues (syringes, white powder etc.).

Without the capacity to form such associations, addiction would not be an issue. Rebecca would fail to remember what actions led to such euphoric experiences, and no motivations would be aroused at the sight of drug paraphernalia. Addiction arises from the often-useful ability to remember what past actions and events resulted in positive and negative experiences. In Rebecca's case, the euphoria experienced while initially using the drug led to a neurological rewiring. It is as if the cognitive system used for goal pursuit has been hijacked to pursue heroin, even in the face of adverse consequences (Panksepp et al., 2002; Tan et al., 2024).

Many aspects of poor mental health, including features of depression, PTSD, and anxiety, only arise because of our ability to learn. For example, anxiety entails worry or concern over some perceived threat (Johnson et al., 2019). The threat has not yet occurred, but to a person with anxiety, past experience of similar circumstances creates excessive fear as they imagine a similar negative outcome occurring in the future. If we can understand what biological mechanisms create associations between a context and a negative outcome, as in the case of anxiety, we can then develop methods which unpick the maladaptive associations and reduce human suffering. While we have made progress in our understanding of how memories are forged at the molecular and cellular level, there are many unknowns which constrain our ability to create useful interventions for maladaptive learning processes (Kandel, 2001; Kandel, 2012; Milton & Everitt, 2012).

Mechanisms of Memory Storage

As described earlier, the process of acquiring an enduring memory involves transforming information from a temporary state to a stable state. But this is not the usual trajectory for most of the information we encounter. The most common fate is for information to be quickly forgotten. This is evolutionarily sensible. Storing information requires energy and biological real estate, and obtaining energy is costly (Jamadar et al., 2025; Karbowski, 2019). Every organism therefore has a small number of things about which it must study extensively and track over time. But most of the information an organism encounters, be it visual, tactile, or olfactory, is not worth storing (Davis & Zhong, 2017). A gazelle cares about the scent of a cheetah, but cares not for the small beetle being crushed underfoot as it moves. While a gazelle's brain may process information about both stimuli, it is not equally likely to store both experiences and the contexts in which they took place. The salience of an object, the meaning it has with respect to our goals and motivations, helps determine what information we retain (Cowan et al., 2021).

When discussing how information is stored, one becomes steeped in complex biological pathways. These pathways involve proteins interacting with other proteins, proteins interacting with DNA, and the production of new proteins. In the neurobiological literature, much of the discussion around learning takes place at the level of the synapse – a synapse being the point where two neurons interact. Typically, the axon from neuron A attaches to a dendrite from neuron B and forms a synapse (Harris & Weinberg, 2012; Montero-Crespo et al., 2020). Synapses are the locus of communication in the brain. At an abstract level, learning occurs when incoming stimulation affects downstream dendritic spines such that they move through the following sequence of stages: generation, stabilisation, consolidation, and maintenance (see Rudy, 2014 for a digestible overview of each stage). Complicated mechanisms are involved in each stage of learning, and not all the components are fully understood. Despite this, the literature provides a good account of many important molecular events thought to be involved in storing information in the brain so that it can be accessed in the future (Kandel et al., 2014).

When some new bit of information has been acquired by the brain, its physical embodiment is referred to as a "memory trace" (Asok et al., 2019; Robins, 2023; Semon, 1921). To generate this memory trace, a postsynaptic influx of calcium (resulting from stimulation by an upstream neuron A) is necessary (Cavazzini et al., 2005; Lynch et al., 1983). Once inside the postsynaptic dendrite, calcium interacts with intracellular proteins that break down actin into smaller chunks (Lynch et al., 2007). Although actin helps give structure to dendritic spines, disassembly is necessary to create room for adding more receptors into the membrane of neuron B. Adding receptors to the membrane makes the spine on neuron B more sensitive to the upstream firing of neuron A, and thus more likely to itself fire an action potential in response. This rapid change in sensitivity (potentiation) is short lived. Without further processing, the potentiated state will revert back to baseline.

Stabilisation of the memory trace requires expanding and strengthening the postsynaptic actin network to solidify the heightened sensitivity (Chen et al., 2007). The spine head is enlarged and, as a result, the actin scaffolding is modified to make it less vulnerable to being disassembled in the future. In addition to actin reorganisation, cell adhesion proteins in the cell membrane help to couple the pre- and postsynaptic neurons, improving the effectiveness of neurotransmission (Huntley et al., 2002). With these two key modifications, the pencil marks of memory are laid down. But these scratchings must be committed to ink to stand the test of time.

Consolidation is where the cellular changes are solidified to stave off forgetting. Consolidation is unique in that it involves the production of new proteins (Frey et al., 1993). In response to neuronal stimulation, a number of events take place in the postsynaptic neuron. One response is the activation of proteins which are able to enter the nucleus and bind to DNA (Martin et al., 1997). This activity leads to transcription of new molecules (messenger RNA) that will later be turned into proteins including receptors. This genomic signalling ensures new proteins are continually minted, providing a sufficient pool of receptors and other elements needed to keep the spine in a state of heightened sensitivity. But the events of consolidation are not final. The postsynaptic cell enters a maintenance stage wherein trafficking of receptors into the membrane is heightened and the typical dynamics of receptor removal and recycling are slowed (Ling et al., 2006; Sacktor, 2011). More excitatory receptors therefore remain on the cell membrane which ensures an enduring change in sensitivity to signals from presynaptic neurons.

Memory Research in Animals and Invertebrates

Many organisms have been poked and prodded in our quest to understand the mechanisms of memory. Scrub jays, a bird sporting a bold blue coat and pointed black beak, have been recurring subjects in studies investigating spatial memory. This is because of their food caching expertise (Shettleworth & Krebs, 1982). Sophisticated techniques have been used to study changes in hippocampal volume in response to caching. The possibility of season-dependent changes in hippocampal neurogenesis in caching birds has also been explored (reviewed in Pravosudov, 2007).

Rodents have featured heavily in the experimental memory literature (Ghafarimoghadam et al., 2022). Recent advances in stimulation and imaging, specifically techniques like optogenetics (Goshen, 2014) and two-photon microscopy (Kawakami et al., 2015), have enabled us to study representation of different types of memory at levels ranging from individual synapses to neuronal ensembles. There is growing interest in understanding the capacity for episodic memory in non-human animals. In the rodent literature, episodic memory is the ability to represent the past and draw on specific encoded events in a manner akin to mental time travel (Crystal, 2022; Eacott & Easton, 2007; Tulving, 2002). Intricate tasks have been developed which enable rats to demonstrate memory for the context in which a stimulus had been previously presented, and to disentangle this from mere familiarity with the stimulus due to temporal proximity (Panoz-Brown et al., 2016). However, the existence of episodic-like memory in non-human animals remains controversial (Hoerl & McCormack, 2019; Tulving, 2005). The establishment of procedures for identifying and manipulating complex episodic memory, by optogenetic and other means, may help identify the mechanisms (synaptic or molecular) that underpin episodic memory in humans.

Research in birds and rodents has supplied decades of insights into the brain regions involved in memory. But to understand the precise structural and molecular changes that underpin the creation of memory, the field turned to invertebrates. The simplicity of the invertebrate neural architecture allows researchers to comprehend and study the entire nervous system. By mapping the connections between all neurons in a given system, researchers can identify the functional specificity of individual neurons. However, simplicity is not the only benefit. Costs can also be significantly reduced, and environmental variables more easily controlled when working with invertebrates. Moreover, ethical concerns are diminished due to the reduced likelihood of finding meaningful sentience at this level.

Although largely unknown outside of the sciences, *Aplysia* is a celebrity among the invertebrates for its contribution to the neurobiology of learning. *Aplysia* is a marine snail with a simple nervous system. The abdominal ganglion (collection of neurons) of *Aplysia* contains the largest known neurons in nature (Moroz, 2011). This makes it an ideal candidate for studies using electrophysiology – the approach where neural activity is recorded by inserting electrodes into cells or in the space surrounding cells. In *Aplysia*, stimulation of the siphon used for transporting water throughout the body leads to a defensive retraction of the gill (Carew et al., 1981). Repeated stimulation leads to a decrease in the intensity and length of the retraction, a simple form of non-associative memory called habituation (Pinsker et al., 1970). Admittedly, this simple form of learning is of limited relevance to human cognition. Yet, this basic adaptation in *Aplysia* served as a platform for understanding the general principles of learning from generation to consolidation and maintenance. It is thought that these stages of information processing are conserved throughout nature and underpin more complex forms of learning that are of interest to humans (Kandel, 2001; Kandel et al., 2021, p. 1330).

Caenorhabditis elegans is another prominent invertebrate which gained widespread attention in the field of neurobiology. This was partly a result of *C. elegans* being the first organisms to have its connectome mapped (White et al., 1986). Understanding the wiring of all 302 neurons in *C. elegans* allowed for a systems perspective of the nervous system. We could piece together the role of each neuron in helping the body to perform actions such as navigation, digestion, and defensive behaviours. Although the nervous system of *C. elegans* is small compared to that of mammals, it revealed principles of neuronal organisation which persist across brains of all sizes. Principles such as reciprocal inhibition to facilitate movement, computing at the level of the cell for efficiency, and minimising the total length of neuronal wire are just as relevant for mammals as they are for invertebrates (Sterling & Laughlin, 2015). We may find comfort in distancing ourselves from so-called “lower organisms”, but nature is indifferent to our need for pre-eminence. Our brains may be bigger, but nature has equipped us with many of the same basic processes for learning, navigating, and operating in a complex world.

The study of invertebrates revealed that even complex behaviour can arise from a modest number of neural cells. Consider that *C. elegans* has less connections in its entire nervous system (~7000) than a single mammalian pyramidal neuron (Cook et al., 2019; Megías et al., 2001; Sterling & Laughlin, 2015). Yet, this bare-bones neuronal setup is sufficient for detecting a variety of chemical and olfactory cues, navigating the environment, escaping threats and detecting dynamic environmental signals such as temperature changes and social crowding. As we climb the ladder of complexity from *C. elegans* to more sophisticated invertebrates, the cognitive capabilities and potential for translational insights expands in turn.

Planaria as a Model Organism

Planaria are a broad group of invertebrates which have become a key part of several areas of research (see Figure 2). Planaria are being used to investigate questions in regenerative biology (Karami et al., 2015), toxicology (Hagstrom et al., 2019; M.-H. Li, 2008), radioprotective materials (Ermakov et al., 2021), addiction (Raffa, 2008), and the effect of zero-gravity environments on morphology (Vista SSEP Mission 11 Team et al., 2018). Planaria have built niches across many ecological contexts and can be found in saltwater and freshwater environments, and also on land. Land dwelling planaria span up to half a metre long (Esser, 1981), whereas freshwater planaria, which are more commonly used in behavioural research, are typically less than a centimetre in length (Vásquez-Doorman et al., 2022).

Planaria are bilaterians. They display bilateral symmetry across their left and right sides (Sluys & Riutort, 2018). Planaria exhibit anterior-posterior polarity, such that their head can be distinguished from the tail in both its structure and its behavioural repertoire. While the tail end of a planarian is rather uninteresting, the head has many intriguing features. Auricles are what give the head of many planarian species a triangular shape. Auricles are thought to support the detection of food and noxious chemicals in the immediate environment (Asano et al., 1998). Eyespots, which are the most discernible feature of planaria, sit atop the dorsal surface of the head. These light-sensitive cell clusters allow planaria to detect light intensity and direction (Shettigar et al., 2017).

Figure 2

Image of two planaria from our breeding colony (species unknown)

Note. There appeared to be two different phenotypes among the planaria collected from the river that supplied our breeding colony (a detailed genetic analysis is ongoing). The left image shows the brown planaria, while the right image shows the darker black planaria. The contrast and brightness of the images were enhanced to make the differences more visible.

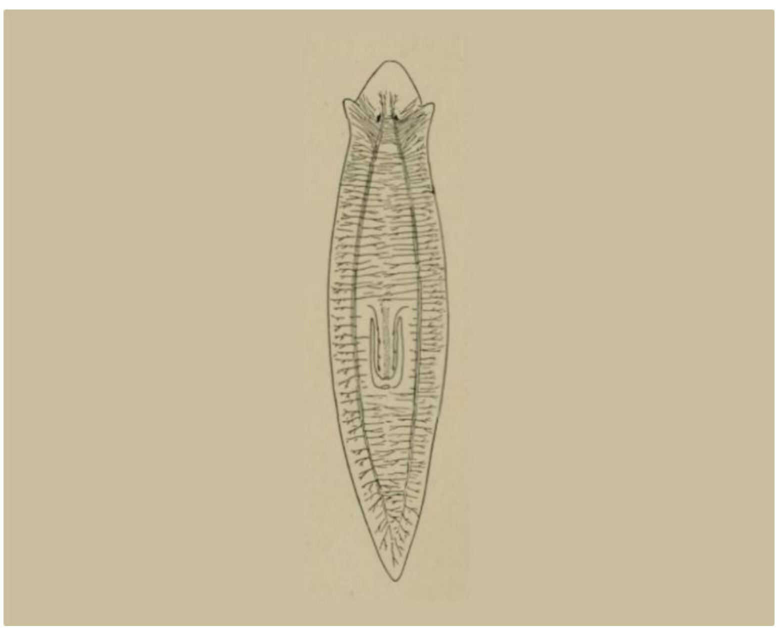
Of particular interest to neuroscientists, the planarian head harbours a bilobed brain which is needed to coordinate activity throughout the body (Inoue et al., 2015). This simple neural structure is of special evolutionary significance as planaria are thought to be the oldest organism to house an organised central nervous system, or what we might call a true brain (Pagán, 2014; Sarnat & Netsky, 1985). In real terms, the planarian brain lacks many features compared to the exuberance of the mammalian brain. But relatively speaking, the brain-to-body-mass ratio of planaria is said to be similar to that of a rat (Best, 1983).

The planarian brain resembles a horseshoe and has been estimated to contain between twenty to thirty thousand neurons (Agata et al., 1998; Inoue, 2017, p. 82; Sarnat & Netsky, 1985). The brain exhibits nine branches on each side which radiate out from the centre. The lobes of the brain form thin nerve cords at their posterior end. These cords extend down the length of the body towards the tail and, together with the brain, comprise the central nervous system (see Figure 3). The left and right nerve cords are connected by commissures which form a ladder-like structure (Sluys & Riutort, 2018).

Planarian neurons are thought to be more similar in structure to those of vertebrates than to those of other invertebrates (Sarnat & Netsky, 1985). They feature spine-like protrusions on their dendrites and contain many dendritic branches but only a single axon (Petralia et al., 2016; Sarnat & Netsky, 1985). Zooming in further, planarian neurons contain a variety of synaptic vesicles, such as clear and dense-core variations, which resemble those seen in vertebrate neurons (Oosaki & Ishii, 1965). Of particular relevance to this project, planaria produce many of the same neurotransmitters and neuromodulators that we humans possess. These include serotonin, dopamine, epinephrine, acetylcholine, GABA, glutamate and opioid peptides (Rawls, Gomez, et al., 2006; Sarnat & Netsky, 1985; Welsh & Williams, 1970; see Buttarelli et al., 2008 for a comprehensive review of planarian neurochemistry).

Figure 3

Anatomy of the planarian central nervous system (species unknown)



Note. Figure 28 from Jordan, D. S. & Heath, H. (1902) Animal Forms; a Second Book of Zoology, public domain. The ventral nerve cords can be seen to run down the length of the body, connected by commissures throughout.

The conservation of neurochemistry demonstrated by planaria is noteworthy, but it is by no means their most interesting characteristic. A more interesting feature relates to the way that planaria reproduce. Many planaria species undergo a natural form of fission as part of their reproductive cycle. They tear themselves in half, with each half then regenerating all the necessary parts of its basic body plan to form a complete planarian again – a form of reproductive immortality. Regeneration is not completely novel in nature. Humans can regrow skin, and salamanders can regrow amputated limbs (McCusker & Gardiner, 2011). But what sets planaria apart from the rest of the natural world is their ability to regrow tissue for the brain and central nervous system. Planaria are able to generate new brain structures de novo and then integrate this into the existing nervous system (Cebrià, 2007).

Planarian regeneration is facilitated by neoblast cells. These are pluripotent adult stem cells which are found throughout the body (Neuhof et al., 2016; Reddien & Alvarado, 2004). After significant injury, these cells proliferate and undergo differentiation, providing the cell types needed to restore organs, membranes, and neural networks in the brain. This process has drawn interest from medical researchers for more than a century (Child, 1941; Morgan, 1898; Reddien, 2018). By understanding the factors that control planarian regeneration, we may make progress towards artificially stimulating these processes in humans to restore lost tissue or even neural structures after injury. Commercial ventures are already being established in this area. Companies with names such as "Morphoceuticals" are looking to apply the lessons learnt from planarian regeneration to rodents and, pending pre-clinical success, eventually humans (Pio-Lopez & Levin, 2023; Saltzman, 2023).

Review of the Literature on Learning in Planaria

In the 1900s, there was still some debate regarding whether invertebrates have the cognitive means needed to learn. A skeptical approach was evident from Donald Jensen in the 1970s who posited that "no invertebrate, no matter how complex is capable of showing 'true learning'" (quoted in Rilling, 1996, p. 591). This view established an artificial barrier separating organisms that suitably model human cognition from those that do not. Because invertebrates were overlooked, researchers tried to make progress on the neurobiology of memory using the complex nervous systems of rodents. After many years searching for the rodent engram – the collection of neurons underlying a specific learning event – it was clear that this venture had unearthed little of value (McConnell, 1967, pp. 2–3). A group of psychologists including James McConnell in the 1970s were aware that little progress was being made in this endeavour. The group moved defiantly away from rodents and drifted towards invertebrates. Using a simple systems approach, basic organisms would allow researchers to progress past mere descriptions and arrive at an actual understanding of the mechanisms of learning.

At first, McConnell and colleagues completed basic experiments showing that planaria could learn to associate a light (conditioned stimulus, CS) with a shock (unconditioned stimulus, US) (McConnell et al., 1959). Compared to control subjects, trained planaria would exhibit more body contractions in response to light and perform more changes of direction. But criticism arose over the lack of controls in these experiments (Travis, 1981). Later follow ups included blinding the experimenter and testing for confounding factors such as pseudo-conditioning, where additional stimuli elicit the unconditioned response despite no temporal relationship. Contrary to the expectations of psychologists at the time, evidence for learning in invertebrates accrued study after study. It was eventually impossible to deny that these rudimentary creatures could form stable associative memories. McConnell, alongside pioneers such as Eric Kandel, amassed definitive evidence that invertebrates are capable of learning, retaining, and acting on information.

Forming associative memories is an impressive feat given the bare-bones layout of the planarian brain. But does this warrant pursuing planaria as a model organism? Why not shift all our resources towards other sophisticated invertebrates like honeybees and fruit flies? It was the pairing of this capacity to learn with the rare ability for regeneration in planaria that sprouted one of the most intriguing branches of research to date: the investigation of memory retention after decapitation and regeneration of the brain. This unique combination allowed researchers to test what happens if you train a planarian and then cut it in half. Would the tail, which must regenerate its head and central nervous system, retain any prior learning? James McConnell, Allan Jacobson and Daniel Kimble were the first scientists to pose and pursue an answer to this question. Across a range of different training procedures, McConnell and colleagues found that regenerated planarian tails indeed retained learned responses (McConnell et al., 1959). This challenged the intuition that memories could only ever be stored in the brain, at least in some instances (McConnell, 1967). It appeared that through some unknown mechanism, memories could be stored or backed up outside the brain and could later be reinstantiated in the new brain during regeneration.

After two decades in the spotlight, interest in planaria eventually waned (Rilling, 1996). This was in part due to a number of controversial studies which tried to investigate the mechanisms underlying the memory retention phenomenon. Thirty years later, this area underwent a modern resurgence thanks to the work of Tal Shomrat and Michael Levin. These researchers published an important paper which used an automated training protocol to revisit the memory retention effect (Shomrat & Levin, 2013). Planaria, like rodents, are hesitant to approach food in the centre of a novel environment (Best, 1963b). They will first explore the territory, and only then engage in consumption. As planaria become familiar with the environment through repeated trials, they begin to approach the food more quickly, demonstrating a form of recognition memory (Best, 1963b). Shomrat and Levin (2013) explored whether this type of memory persists in the tails of trained planaria following complete regeneration of the brain.

Over ten consecutive days, half of the planaria were fed on the novel rough surface ("familiar" planaria) while the other half ("naive" planaria) were only fed on a smooth surface resembling that of the home tank. At the end of the training period, the familiar group took a significantly shorter amount of time to approach and consume the food in the rough environment. Both groups were then bisected into head and tail halves and left to regenerate for 10-14 days. The authors then looked at whether the tail regenerates of familiar planaria retained familiarity of the rough environment and thus approached food more quickly compared to the naive tail offspring. The data revealed that regenerated tail fragments from familiar planaria did approach the food more quickly, however, this did not reach statistical significance. After undergoing the same training procedure as the original planaria, the authors found that regenerated tail fragments from familiar planaria demonstrated a form of memory savings. The familiar tail regenerates became accustomed to the rough environment faster than regenerates of control planaria. This indicated that some memory trace from prior training survived brain regeneration but required repetition of the training process for the memory savings to be expressed.

More recently, Samuel et al. (2021) corroborated this puzzling memory retention effect. The authors used sucrose to shift the surface preference of planaria from their typical preference for a smooth surface to the sucrose-paired rough surface. After amputating the planaria and allowing time for head regeneration, it was observed that the tail halves retained the sucrose-paired rough preference, despite the newly regenerated brain never having been exposed to the rough surface. In contrast, the tail halves of control planaria – which were exposed to the rough surface but did not receive sucrose in this environment – showed the usual preference for the smooth surface.

In the memory retention experiments above, it was presupposed that although a brain is not necessary for memory storage, it is needed to act upon the memories. For this reason, sufficient time was given to allow the brain to regenerate. However, a recent preprint by Shimojo et al. (2022) challenges this assumption. They tested whether planarian tails can show retention of a conditioned response prior to regeneration of the brain. In this study, planaria were trained to associate a neutral weak UV light (conditioned stimulus) with an aversive shock (unconditioned stimulus). The authors described that a shock typically causes planaria to twist their body – an unconditioned contortion response. After pairing the light with the shock, planaria displayed a conditioned contortion response to the UV light alone.

After the response was established, the authors cut the planaria into head and tail halves. On the second and third day after dissection, before brain regeneration is thought to be complete (Roberts-Galbraith et al., 2016), the tail halves were exposed to the conditioned stimulus. The authors found that most responses from the tail halves resembled those produced by an electric shock rather than those produced by a neutral ultraviolet light. Ultimately, this suggested the tail halves retained the conditioned behaviour and were able to act on it despite lacking a complete brain at the time.

Rhodes and Vierick (2024) followed a similar procedure to establish negative phototaxis in planaria (moving away from light). Typically, planaria are strongly averse to blue light, mildly averse to green light, and are indifferent to red light (Paskin et al., 2014). Planaria were trained to associate a neutral red light with an aversive green light across 5 days. After conditioning, half of the planaria were bisected into head and tail halves and the rest were left intact. Three weeks later, all planaria were tested for retention of the conditioned response. Both head and tail regenerates retained the conditioned response as well as intact planaria. Moreover, memory retention was not statistically different when comparing head regenerates to tail regenerates. This study adds to the evidence suggesting that tail regenerates can retain and act on a memory even after total loss of the brain.

Admittedly, there were a number of issues with the study by Rhodes and Vierick (2024). These represent common limitations found in the planarian literature. First, the number of planaria per group was low (between four to six subjects). Second, it was not clear how the dependent variable was operationalised. For example, how much movement was necessary to qualify as negative phototaxis on a given trial? We must maintain skepticism for individual studies given their limitations. But the number of findings showing successful retention of learning through regeneration provides strong support for the phenomenon.

Classical conditioning procedures are most common in the planarian literature. But some experimenters have also employed operant conditioning methods (Chicas-Mosier & Abramson, 2015; Crawford & Skeen, 1967; see Best, 1963a for a review of early studies). A simple learning procedure known as the Van Oye maze was one of the first forms of reinforcement learning in planaria (Nicolas et al., 2008; Oye, 1920; Wells, 1967). In the typical setup, planaria are housed in a beaker and a fishing line with food is suspended just below the water surface. Planaria can detect the presence of food and navigate towards it (Ash et al., 1973; Miyamoto & Shimozawa, 1985). To reach the food, planaria must navigate up the wall, across the surface and down the fishing line. This is a low probability behaviour, but some small percentage of subjects will find their way to the fishing line and be reinforced by the food.

When the Van Oye procedure is used, many planaria learn to perform the set of movements needed to reach the food when it is present in the beaker (Wells, 1967). Importantly, control planaria undergo the same procedure but without the food reward attached. At test, food is not placed on the rod and is instead dissolved in the water beforehand. The dissolved food is a cue that food is available. During the test, trained planaria are subsequently found in much greater numbers on the suspended line compared to control subjects. Across five experiments performed by Wells, an average of ~17 trained subjects were found on the line at test compared to an average of ~3 experimental subjects (reviewed in Corning & Riccio, 1970). This procedure demonstrated that planaria can be trained using reinforcement learning.

Another operant conditioning study was conducted by Corning (1966) during the height of planaria fame. Corning wondered whether operant conditioned behaviours could persist through regeneration. Using a T-shaped apparatus, planaria were trained via positive reinforcement to select their least preferred side. Reinforcement consisted of being returned to the home arena for 10 minutes after making a correct choice. After incorrect choices planaria were taken to the start of the maze for another trial. A threshold for successful learning was set at nine out of ten consecutive correct choices across trials. Planaria that met this threshold were bisected.

After a two-to-three-week regeneration period, the regenerates (both heads and tails) were given a baseline preference test and were subsequently conditioned to criterion. Corning found that the baseline preference (left or right) of trained tail regenerates resembled the conditioned preference of the original planaria before bisection. Tail regenerates of untrained planaria instead resembled the baseline preference. This suggested that the trained tail regenerates retained the prior learned preference, implying that operant conditioned behaviour can be retained outside of the planarian brain. Furthermore, the regenerates of trained planaria could also be conditioned to threshold faster than regenerates of untrained planaria. While this provided evidence of memory savings when re-exposed, it also demonstrated a form of uncued recall of the memory.

Building on the earlier work described above, Read (2021) investigated whether a Y-shaped maze can be used to shape a directional preference in planaria. In Experiment two of Read's paper, baseline directional preferences were obtained for planaria by allowing them to complete six trials in the Y-maze and recording whether they entered the left or right arm more frequently. The planaria then underwent a conditioning procedure such that they were rewarded with 2% ethanol if they entered their non-preferred arm ("active arm"). On day four of conditioning, planaria entered the active arm significantly more often than during baseline. This result supports the conclusions of Corning (1966), suggesting that planaria may be capable of learning a directional preference in a Y-maze. Unfortunately, the behaviour was only significantly different from baseline on the final day of conditioning (day four). It is therefore not clear whether this conditioned response was stable or the result of chance variation. Optional stopping may have increased the chance of a false-positive finding within this study, as the number of conditioning days differed between experiments.

A related study investigated the ability for planaria to be conditioned in a Y-maze using cocaine as the reinforcing agent (unpublished data, Canales laboratory). After identifying baseline preference of each subject, the non-preferred arm was reinforced with cocaine across three conditioning days, with three trials each day. Cocaine treated planaria showed a rapid increase in active arm entries, choosing the cocaine reinforced arm more than 90% of the time across the final two days of conditioning. This effect was replicated among a separate cohort of planaria. Other groups were also run which received cocaine in conjunction with different doses of either ceftriaxone or N-acetylcysteine; these drugs reduce addiction-like behaviour in rodent models (Baker et al., 2003; Sari et al., 2011). Unfortunately, there was no vehicle only group. Nevertheless, Planaria treated with ceftriaxone or N-acetylcysteine alongside cocaine did not acquire a conditioned response.

The experiments above provide preliminary evidence that planaria can learn an operantly conditioned response. But when considering whether this behaviour can persist through bisection and regeneration, there is very limited evidence. Moreover, much of the research on operant conditioning in planaria dates back to the mid-twentieth century. Although historical research still holds value, modern psychological science has raised questions regarding the reliability and replicability of past experiments.

Recent evidence suggests that the psychological literature broadly considered has oversold the robustness of many psychological phenomena (Open Science Collaboration, 2015). Much effort is being devoted towards identifying the types of decisions which led to unreliable results appearing in the literature in the first place (Simmons et al., 2011). Interestingly, many scientists openly admit that they have engaged in questionable research practices – design and analysis decisions which led to untrustworthy results that fail to replicate (Gopalakrishna et al., 2022). Although the problem of replicability focuses on findings from the last two decades, this general trend should generate skepticism when considering research from the twentieth century. Especially in cases where there are only one or two reports of a given phenomenon. For this reason, we must seek to establish reliable methods for inducing operant conditioned behaviours in planaria. Furthermore, we should withhold judgement on whether learned behaviours can persist through decapitation and regeneration in planaria until the phenomenon reported by McConnell et al. (1959) and others is replicated.

Cocaine-Induced Behavioural Plasticity in Planaria

Investigators have used many different stimuli, both aversive and appetitive, in their efforts to condition planaria. Cocaine, which acts primarily on the dopamine transporter, is one of the most common appetitive stimuli used to date. Importantly, these transporters are abundant in planaria (Algeri et al., 1983; Buttarelli et al., 2008). The abundance of dopamine transporters, combined with the small quantity of cocaine needed to reward such a small organism, makes cocaine a cost-effective tool for conditioning experiments. That said, there are some concerns that require consideration when administering cocaine in behavioural tasks. For example, cocaine induces strong effects on locomotion and atypical behaviours at some doses (Pagán et al., 2013; Rawls et al., 2010).

Cocaine exerts its agonistic effects by blocking reuptake of dopamine through the dopamine transporters. In humans, this results in more dopamine activity in the synapse and therefore altered neural activity in downstream neurons, particularly in the meso-limbic pathway connecting the ventral tegmental area to the nucleus accumbens (Nestler & Lüscher, 2019). This agonistic effect is linked to the high that cocaine users experience, an effect shared by all drugs of abuse (Pierce & Kumaresan, 2006). Cocaine also acts on serotonergic and noradrenergic transmission by blocking their respective transporters (Galli et al., 1995; Mateo et al., 2004). The noradrenergic effects are thought to stimulate the sympathetic nervous system by blocking reuptake of noradrenaline and decreasing sympathetic nerve discharge, resulting in effects such as increased blood pressure and heart rate (Freye, 2009; Jacobsen et al., 1997; Nestler & Lüscher, 2019). The pharmacokinetics of cocaine activity in planaria is only partly understood. Yet, the presence of all three of these neuromodulators in planaria make it an ideal candidate for behavioural reinforcement experiments (Buttarelli et al., 2008).

Amaning-Kwarteng et al. (2017) explored the establishment and extinction of a cocaine-reinforced texture preference. They found that planaria can be conditioned using cocaine to shift their surface texture preference from smooth to rough and that this preference can be extinguished after repeated trials without reinforcement. Subsequently, exposure to a bath of cocaine was enough to reinstate the conditioned preference when given free access to both surfaces.

Building on prior work dating back to the 1960's (Needleman, 1967), Mohammed Jawad et al. (2018) investigated addiction-like behaviour in planaria through conditioning, extinction and tolerance. The experiment successfully demonstrated a conditioned place preference (CPP), extinction of the preference, and context specific tolerance. Of particular significance, this work demonstrated that sucrose induced CPP requires dopaminergic activity. Administration of a dopamine D1 antagonist during conditioning blocked acquisition of CPP but did not interfere with context specific tolerance. An interesting dissociation which the authors suggest may have implications for understanding addiction in humans.

The studies described above suggest that the behaviour and preferences of planaria can be successfully shaped using cocaine. Despite the long evolutionary distance between rodents and planaria, it is interesting that their drug seeking behaviour resembles that of drug dependent rodents. In particular, the associations acquired during learning can be brought back upon exposure to the drug, like cue induced reinstatement in rodents. While interesting, the training procedures typically used for planaria are rudimentary. Simple associative learning paradigms fail to capture important aspects of drug seeking in animals, like the complex tasks performed to gain access to the drug (Singer et al., 2018), or the pursuit of drugs in the face of negative consequences. To generate findings with a greater chance of generalising across the evolutionary tree, it would be advisable for the field of planaria research to establish conditioning procedures that more closely mimic those in mammals.

Unresolved Questions

In the first half of the 20th century, there was doubt regarding whether invertebrates can learn. But as we look back nearly a century later, we have gathered ample evidence that planaria and many other invertebrates can form long-lasting memories (Amaning-Kwarteng et al., 2017; Samuel et al., 2021; Wells, 1967). Planaria are an especially useful organism given their ability to learn and their unique ability to regenerate. As has been shown with conditioning procedures, there is now evidence that memory can be successfully retained outside of the brain (Shomrat & Levin, 2013). The persistence of basic associative memory through regeneration is remarkable.

But a more compelling finding that would truly shake our fundamental understanding of memory storage mechanisms would be the persistence of complex behavioural responses.

An acquired texture preference is a valid form of learning. But it is far removed from the memories that concern us in our day-to-day lives. In contrast, learning shaped by a reward better reflects the intentional learning we associate with intelligence and meaningful behaviour in humans. If complex memories formed by operant conditioning can persist in planaria despite complete loss of the brain, this may have profound implications for the way we view memory storage and retrieval in humans. This project aims to test whether memory retention through regeneration, previously shown for classical conditioning, also occurs for operant conditioning.

The results of Shomrat and Levin (2013) suggested the memory trace lay dormant and failed to be reactivated at first. After exposure to the same training procedure that led to the original memory formation, the dormant trace was then reawakened. This phenomenon of memory reactivation after prior failures parallels behaviour reinstatement in addiction research. After successfully training an animal to lever press for a reward such as cocaine, the lever press response can be extinguished by allowing the animal to repeatedly engage in the behaviour without being rewarded (Wit & Stewart, 1981). Eventually, the animal will stop performing the conditioned response when the lever is presented. However, if the animal is exposed to the reinforcing stimulus before being placed back in the operant chamber, the lever pressing behaviour will spontaneously return (Wit & Stewart, 1981).

With respect to memory savings in planaria and memory reinstatement in rodents, the memory is either temporarily inaccessible or actively inhibited, and requires exposure to the right stimulus to be reactivated. Although extinction and reinstatement of drug seeking behaviour has been modelled in planaria (Amaning-Kwarteng et al., 2017), no experiments have explored whether a reinstatement procedure can also be used to reactivate memories which may remain dormant after decapitation and regeneration. The phenomenon of memory savings among regenerates found in Shomrat and Levin (2013) demonstrated that some memory trace is retained in the brainless tail half. Perhaps this trace can be reactivated without the need for retraining by instead exposing the tail regenerates to the reinforcing compound. This project will therefore investigate whether memory stored outside the brain behaves like an extinguished memory such that exposure to the reinforcing compound is sufficient to reinstate the memory trace.

Experiment 1

Experiment 1 aimed to find a dose of cocaine that would not significantly alter the locomotor behaviour of planaria. This was important to ensure the rewarding compound did not inhibit the ability for planaria to respond during subsequent trials in the Y-maze experiments reported below. To ensure that our selected dose was rewarding despite a lack of effect on movement, we sampled a range of doses previously used for conditioning in the literature. During classical conditioning procedures, cocaine has been used in concentrations ranging from 1 µM (Hutchinson et al., 2015) to 80 µM (Raffa et al., 2005) to shape texture or light preferences.

Although cocaine has been used many times to study addiction-like behaviour, several researchers have observed that some planaria species are more amenable to conditioning procedures than others (Mueller & Levin, 2002; Samuel et al., 2021). Moreover, differences in the behaviour and responses of planarian species have been observed for several types of stimuli (Cochet-Escartin et al., 2015; DeBold et al., 1965). For this reason, it was important to perform a dose-response analysis to identify a concentration of cocaine which does not significantly alter motility in the planaria species used for this project.

Colony Maintenance and Handling

Due to restrictions on importing identified species such as *Schmidtea mediterranea* into New Zealand, planaria were sourced from a local stream in Wellington, New Zealand. Given the basic characteristics of the planaria we collected (colour, head shape etc.), it was suspected that they were *Neppia*, a species commonly found in New Zealand waterways. We are in the process of performing species level identification. But at the time of publication, the species had not been identified.

Planaria were housed in a 50-litre glass aquarium with internal filtering. The aquarium contained a natural ecological environment (rocks, snails, algae etc.). The tank water (referred to as "planaria water" hereafter) was maintained with Prime – a concentrated water conditioner. Planaria were fed between one and three times a week, with meals consisting of frozen liver paste. The colony was maintained on a 12-hour light/dark cycle with lights on at 9:30am till

9:30pm. For this experiment, planaria were handled using either a filbert (medium length flat) paintbrush or a fine artist's paintbrush.

Materials and Procedure

Plastic petri dishes with a diameter of 5.5 cm were used to assess motility. Planaria locomotion was captured using an OPPO A17 smartphone and the videos were imported into EthoVision (Noldus Information Technologies, Wageningen, the Netherlands) for motility tracking. When subjects were not visible due to being occluded by a shadow or dish wall (~10% of frames on average across all subjects), missing data were interpolated using the interpolation feature in EthoVision. This imposed a direct line from the subject's last location to the next observed location to determine the distance travelled.

Sixty planaria were used in this experiment, with subjects evenly distributed across six conditions. The conditions corresponded to concentrations of cocaine commonly used in the literature: 0, 1, 5, 10, 20 and 100 µM (*n* = 10 per condition). This experiment had twelve recording sessions[1]. Each subject participated in one recording session, and five subjects were run in each session. Sessions lasted 15 minutes. Subjects were collected from the breeding tank on the day of data collection. Planaria length was not measured, but only those visually estimated to be between 0.5 cm and 1 cm were selected for this experiment. Within each session, subjects were randomly allocated to their condition using a freely available random number generator.

The desired drug concentrations were achieved by diluting a 250 µM cocaine hydrochloride stock solution (BDG Synthesis, Wellington, New Zealand; prepared in distilled water) with planaria water to a final volume of 8 ml. All drug solutions were prepared prior to the first recording session of the day and were mixed before use. For each session, planaria were picked up at random and a randomly generated number sequence was used to determine which condition it was assigned to. The recording began once all five subjects were in their respective

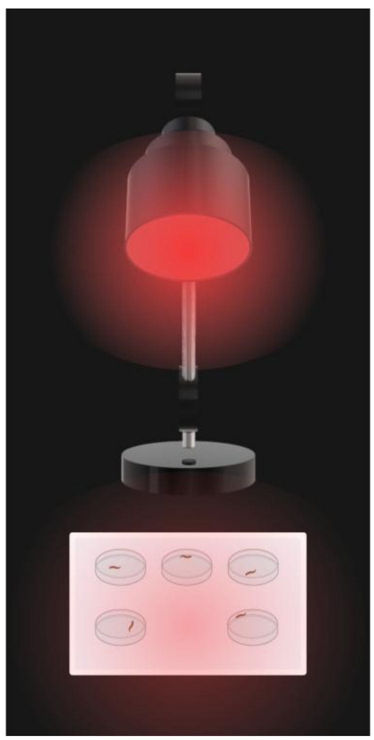
[1] The first ten sessions each contained one subject from the 0 – 20 µM conditions, whereas the last two session only contained subjects in the 100 µM condition. This is because the 100 µM condition was added after the initial data were analysed to ensure that the cocaine could have an effect on the planaria and was not inert.

dishes. After completing a single trial, planaria were rehoused in a large tank and were not used for any subsequent experiments in this manuscript.

Figure 4 shows the recording setup used for this experiment. Five petri dishes were positioned on a white acrylic sheet. Recording sessions took place under red light, with the light positioned 36 cm above the dishes. The dishes were aligned in a 2x3 grid, with a gap left in the middle position. The overhead light was centred in the gap to minimise shadows cast over the dishes – this was important to facilitate accurate tracking in EthoVision. Each drug concentration was rotated through the 5 grid positions across trials to control for any effects of lighting angle.

Figure 4

Graphical Depiction of the Dose-response Apparatus



Note. Graphical depiction of the dose-response setup used to assess planarian motility in response to cocaine. The room was illuminated by a lamp fitted with red plastic to filter out non-red light. All other lights were turned off during data collection. The arrangement of petri dishes and planaria on the white plastic sheet can be seen below the lamp. An Oppo A17 phone was used to record planaria motility (not shown in the graphic). The phone was balanced on a clamp attached to a support stand positioned to the side of the plastic sheet.

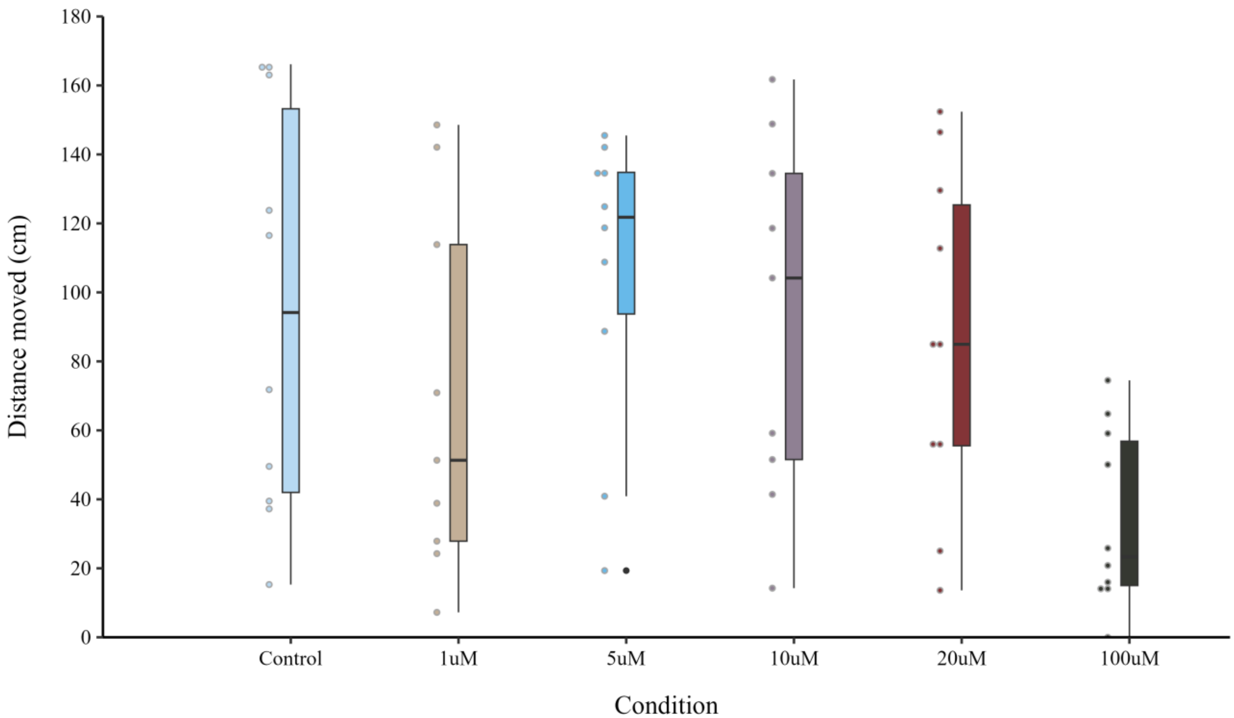
Results and Discussion

Figure 5 depicts the distance moved by planaria across the six conditions. Prior to performing any statistics, the assumptions of normality and homogeneity of variances were tested. Levene's test for homogeneity of variances suggested there were equal variances across conditions ($F = 1.95$, $p = .101$). The Shapiro-Wilk test indicated that the data were not normally distributed ($W = 0.935$, $p = .003$). Due to violation of the assumptions of ANOVA, a Kruskal-Wallis test was used to evaluate group differences. The results show a statistically significant effect of condition on distance moved (χ^2 (5) = 11.3, $p = .045$). An exploratory post-hoc Dunn's test was carried out to determine the group differences. The results indicated that the 100 μM group differed significantly from several other groups: control ($p = .006$), 5 μM ($p = .002$), 10 μM ($p = .003$), and 20 μM ($p = .016$). No other significant differences were found.

The results in Figure 5 convey the variability of planarian behaviour. All conditions had at least one subject which moved less than 30 cm over the 15-minute recording, and all groups had at least two subjects that moved more than 140 cm. The experimenter observed that when placed in the recording dish, some planaria would move initially, and then come to rest within a few minutes at a spot on the wall. They would remain there without meaningful movement for the remainder of the recording. Although the 1 μM group did not differ significantly from the control group, there was a curious grouping of planaria in the 1 μM condition clustered below 60 cm. Consistent with this, Hutchinson et al. (2015) observed a significant decrease in motility during exposure to 1 μM, but not to 10 μM, of cocaine when compared to controls. No potential explanation was offered for this unusual curvilinear pattern.

Figure 5

Median Planaria Motility For Each Condition



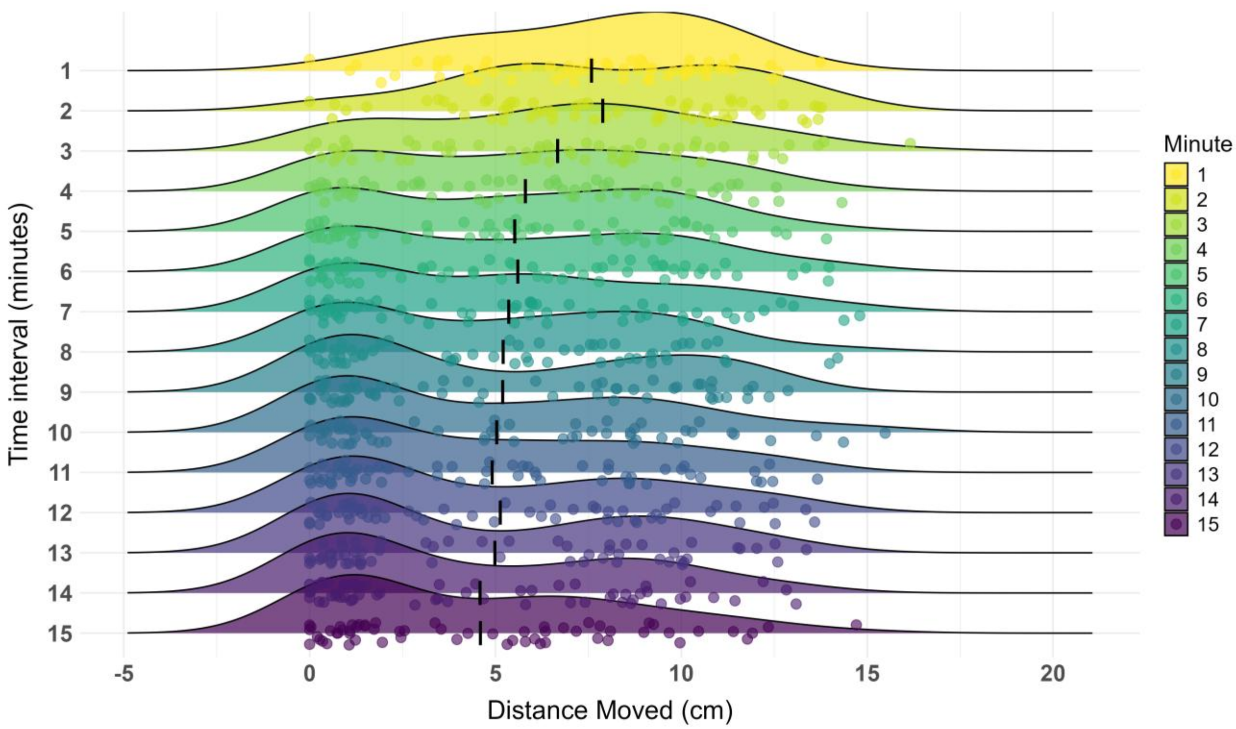
Note. Box and whisker plot of distance moved by planaria over the 15-minute recording interval. Black bars indicate the median distance moved for each condition. The boxes represent the Interquartile range (IQR) with whiskers extending 1.5 times the IQR. The 100 µM group differed significantly from the control, 5, 10, and 20 µM groups, with the 100 µM group covering less distance in each case.

The results suggest any dose between 1 µM and 20 µM could be used for conditioning without systematically affecting planaria motility. It was also necessary to select a dose that was sufficiently rewarding for planaria. A range of doses have been used to successfully condition planaria in CPP paradigms. These procedures typically involve low doses such as 1µM (Hutchinson et al., 2015), 5 µM (Amaning-Kwarteng et al., 2017) or 10 µM (Hutchinson et al., 2015). It is worth noting that in the case of CPP, drug exposure time is relatively long. On the order of 15-20 minutes per trial. Whereas in the operant conditioning paradigm proposed in Experiment 2 of this project, exposure time would be just 3 minutes. To adjust for the smaller absorption window compared to CPP experiments, the larger 20 µM concentration was chosen for experiment 2.

Alongside total motility, we were able to inspect how planaria motility changed over the recording interval. We observed that planaria moved more at the start of the session compared to the end, with a gradual decrease in the distance moved with each passing minute (see Figure 6). An exploratory Welch's two sample t-test found a significant difference between the distance travelled in the first minute ($M = 7.58$, $SD = 3.29$) compared to the 15th minute ($M = 4.59$, $SD = 3.95$), with subjects travelling significantly further during the first minute ($t(59) = 6.2$, $p < .001$). In future dose-response assessments, shorter recording sessions may suffice to assess dose-response curves.

Figure 6

Planarian Motility Across the Recording Session



Note. Ridge plot of distance moved by planaria during each minute interval. Each ridge shows the distribution of distance moved for all subjects during the minute interval. Black bars indicate the mean distance moved for the whole sample (treatment and control subjects).

Experiment 2

Prior research has demonstrated the capacity for learning in planaria by way of classical conditioning. Moreover, it has been further shown that classically conditioned memories can be retained after decapitation and regeneration of the brain (Samuel et al., 2021; Shomrat & Levin, 2013). But the capacity for complex memories shaped by operant conditioning to persist under these conditions has not been definitively shown. As a first step towards assessing whether operantly conditioned memories can persist through decapitation, we must first demonstrate the capacity for operant learning in this species of planaria. The power analysis, experimental design and analysis plan of this experiment were preregistered prior to data collection. The preregistration can be found online at Open Science Framework and at PsycArchives.

Colony Maintenance and Handling

The colony maintenance protocols were identical to those described in Experiment 1. To track subjects throughout the experiment, planaria were housed individually in 12-well plates with 2 ml of planarian water (changed daily[2]). Planaria length was not measured, but only those visually estimated to be between 0.5 cm and 1 cm were selected for this experiment. Planaria were stored in a room illuminated with standard white fluorescent lighting on a 12-hour light/dark cycle with lights on from 9:30am till 9:30pm. Subjects were moved into a room dimly illuminated with red light while completing their Y-maze trials.

Planaria were handled using different techniques for different circumstances. When removing planaria from their 12-well plate, a filbert (medium length flat) paintbrush was preferred. However, when moving planaria between petri dishes and the Y-maze, a fine artist's paintbrush was preferred. In other cases, such as when planaria would sit in the middle of the Y-maze divot, a plastic transfer pipette with the tip cut off was used. Planaria were gently handled

[2] In the waiting period between conditioning and the memory retention test, 12 subjects (#49 - #60 inclusive) from were left overnight without water due to experimenter error. This led to three immediate deaths. Although the remaining subects survived, during the memory retention and reinstatement tests, 6 failed to respond at all and the remaining 3 subjects only responded once or twice.

throughout their lifespan. Rough handling was suspected to have caused a high mortality rate during pilot experiments.

Materials and Procedure

This experiment used two groups: a treatment group (*n* = 30) which received cocaine and a control group (*n* = 30) which received vehicle only (distilled water). There were four experimental stages: baseline, conditioning, test, and reinstatement (see Figure 7). A modified version of the Y-maze conditioning procedure outlined by Read (2021) was adopted. During baseline and conditioning trials two planaria were run concurrently in separate Y-mazes (see Figure 8). Each maze was filled with 1.8 ml of planaria water which was shaken gently to evenly distribute the water throughout the runway and arms. Six planaria were used per run, wherein they completed either six (baseline) or four trials per day (conditioning) with an intertrial interval of approximately 15 minutes.

At the start of each run six planaria were moved into petri dishes. At the start of a trial, planaria were transferred to the middle of a maze runway using a paintbrush. Each maze contained one planarian. A timer was started once each planarian was placed in the runway. Planaria were given three minutes to enter one of the arms[3]. Once a subject entered an arm, the plug was inserted to stop liquid moving between compartments, after which 0.5 ml remained in each arm[4]. A planarian was considered to have entered the arm when the plug could be safely inserted without touching the planarian.

When treatment subjects entered the active arm, 43.5 µL of cocaine (BDG Synthesis, Wellington, New Zealand) in distilled water was pipetted near the centre of the arm to achieve a 20 µM concentration. If the inactive arm was selected, an identical volume of distilled water was pipetted near the centre of the arm. After administration, the timer was restarted and three minutes were given for absorption. For control subjects, entry into either arm resulted in 43.5 µL of distilled water being pipetted into the arm. If a subject failed to enter an arm, the plug was

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[3] If the planarian had some part of their body in an arm at the end of the allotted time, they would be given up to an extra minute to make their decision.
[4] Measured by extracting and weighing the volume of distilled water trapped in each arm, assuming a density of 1 g/ml. Small deviations are inevitable, but this was the most common value seen during our tests.

inserted and 43.5 μL of distilled water was pipetted into the runway and then three minutes were given. The runway light was on throughout the duration of the trial. At the end of a trial, planaria were gently removed and placed back into their holding dish. Mazes were rinsed and dried between each trial.

The memory retention test took place 14 days after conditioning (see Figure 7). At test, six planaria were used per run. Three planaria were run concurrently in three separate Y-mazes. Planaria were given three minutes to enter an arm. Once a decision was made, the plug was inserted and planaria were left for approximately 60 seconds before being moved back to the holding dish. No additional liquid was added during test trials. The next group of three planaria would then begin their first test trial. The inter-trial interval was approximately six minutes and thirty seconds. A reinstatement procedure was carried out the following day. The procedure was identical to the test stage with the only additional step being drug exposure before the first trial. At the start of a run, planaria were placed individually into an 8 ml solution of planaria water containing 20 μM of cocaine for 10 minutes. At the end of the exposure interval, planaria were moved into separate Y-mazes to begin their first trial. Planaria were only exposed to cocaine prior to the first reinstatement trial.

There were three exclusion criteria identified in the preregistration document. The exclusion criteria were: A) failing to complete at least four of the six baseline trials; B) failing to complete at least two trials on consecutive conditioning days; C) failing to complete at least four of the last six trials of conditioning. We attempted to replace all subjects that met criterion A. However, due to time constraints, of the 18 that failed to meet this criterion, only 13 could be successfully replaced. Five subjects could not be replaced and so started conditioning despite having completed only two or three baseline trials. Thirteen subjects met criterion B or C and were excluded from the data analysis. There was no exclusion criteria set for the test and reinstatement procedure. However, some subjects died in the waiting period, or demonstrated greatly impaired behaviour at test or reinstatement, and were thus excluded.

Three laser-etched Y-mazes were used for this experiment (see Figure 8 for dimensions). Mazes were laser etched into 80x80 mm plastic squares. At the intersection between the runway and the arms, there was a small divot on the floor of the maze. This allows a plug to be inserted to trap liquid in the arms and enable controlled drug administration. The maze floor contained subtle lines as a result of the etching process. At the base of the runway there was a small externally powered white light (~20 lux) which was fixed into the plastic. Light is an aversive stimulus which induces negative phototaxis and should discourage planaria from resting at the start of the runway.

Figure 7

Graphical Timeline of Experiment 2

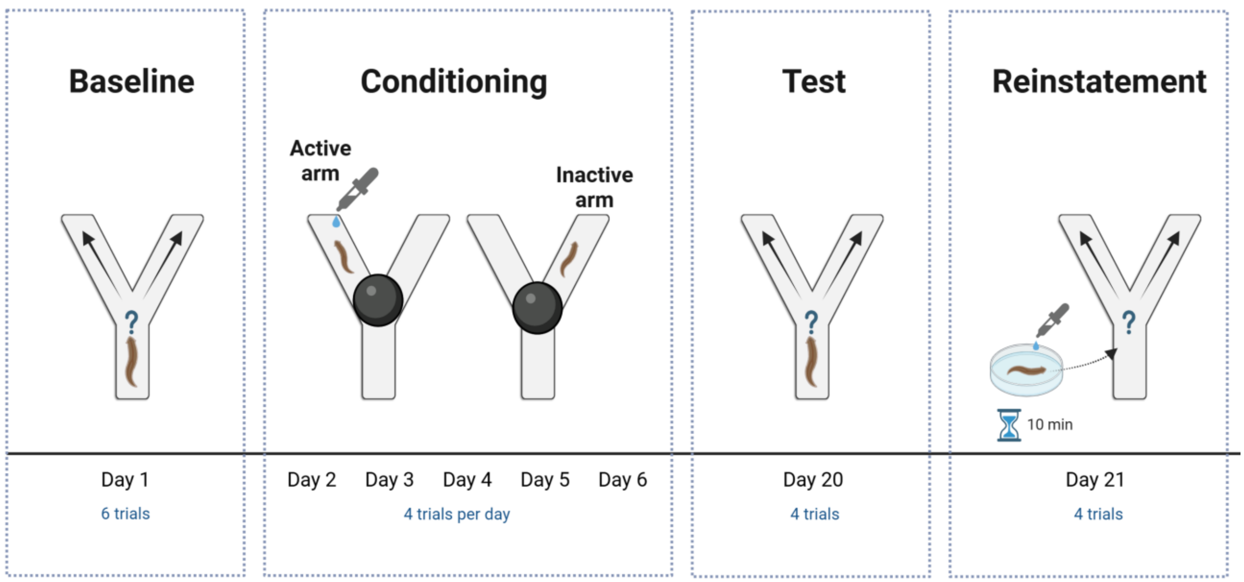


Figure 8

Laser Etched Plastic Y-maze

Note. The Y-maze depicted here was laser etched into a white 80x80 mm plastic plate. A white LED was embedded in the maze at the start of the runway to induce negative phototaxis (light bulb symbol is indicative of LED location beneath the plastic). The light was powered by a 9V power adapter. A plastic plug was also etched out of plastic to stop liquid moving between compartments after a planarian entered one of the arms.

Results and Discussion

Of the 60 original subjects, 3 died during conditioning (2 control subjects and 1 treatment subject) and another 6 subjects died throughout the two-week waiting period (5 control subjects and 1 treatment subject). The initial deaths were attributed to repeated handling with a paintbrush, while the deaths during the waiting period were in part due to 12 subjects being left overnight with no water. Additionally, 10 other subjects were excluded due to meeting one or more of the exclusion criteria during conditioning. Of the subjects excluded due to death or meeting the exclusion criteria, 9 were control subjects and 4 were treatment subjects. Due to different requirements for the between-groups and within-subjects comparisons for active arm entries, and due to some subjects having no data for the decision latency analysis, there were different numbers of subjects for the comparisons at each time point. For this reason, the number of subjects per condition at each time point have been included in the figures shown below.

After removing subjects that met the exclusion criteria, the majority of subjects had an initial preference towards the right arm (n = 25), with just over a quarter favouring the left arm (n = 13) and several having no preference (n = 9). This experiment employed a biased design, such that the active arm to be reinforced was the opposite of the initial preference or randomly assigned for those with no initial preference. The left arm was active for 28 subjects, and the right arm was active for 19 subjects. Looking at the baseline preferences of excluded subjects, there was no systematic bias towards either arm, with slightly more preferring the right arm (n = 6) than the left arm (n = 4), and a few showing no preference (n = 3).

Figure 9 shows the average proportion of trials for which subjects entered the active arm at each time point. A generalised linear mixed effects model with family set to binomial was fitted in R using the lme4 package (Bates et al., 2015). Subject ID was set as a random effect, with condition, time point and the interaction term as fixed effects. Pairwise comparisons with a Bonferroni correction were carried out using the emmeans package in R (Lenth, 2024). A Type III ANOVA was conducted using the car package (Fox & Weisberg, 2019) to identify whether there was a significant effect of condition or time, or an interaction effect.

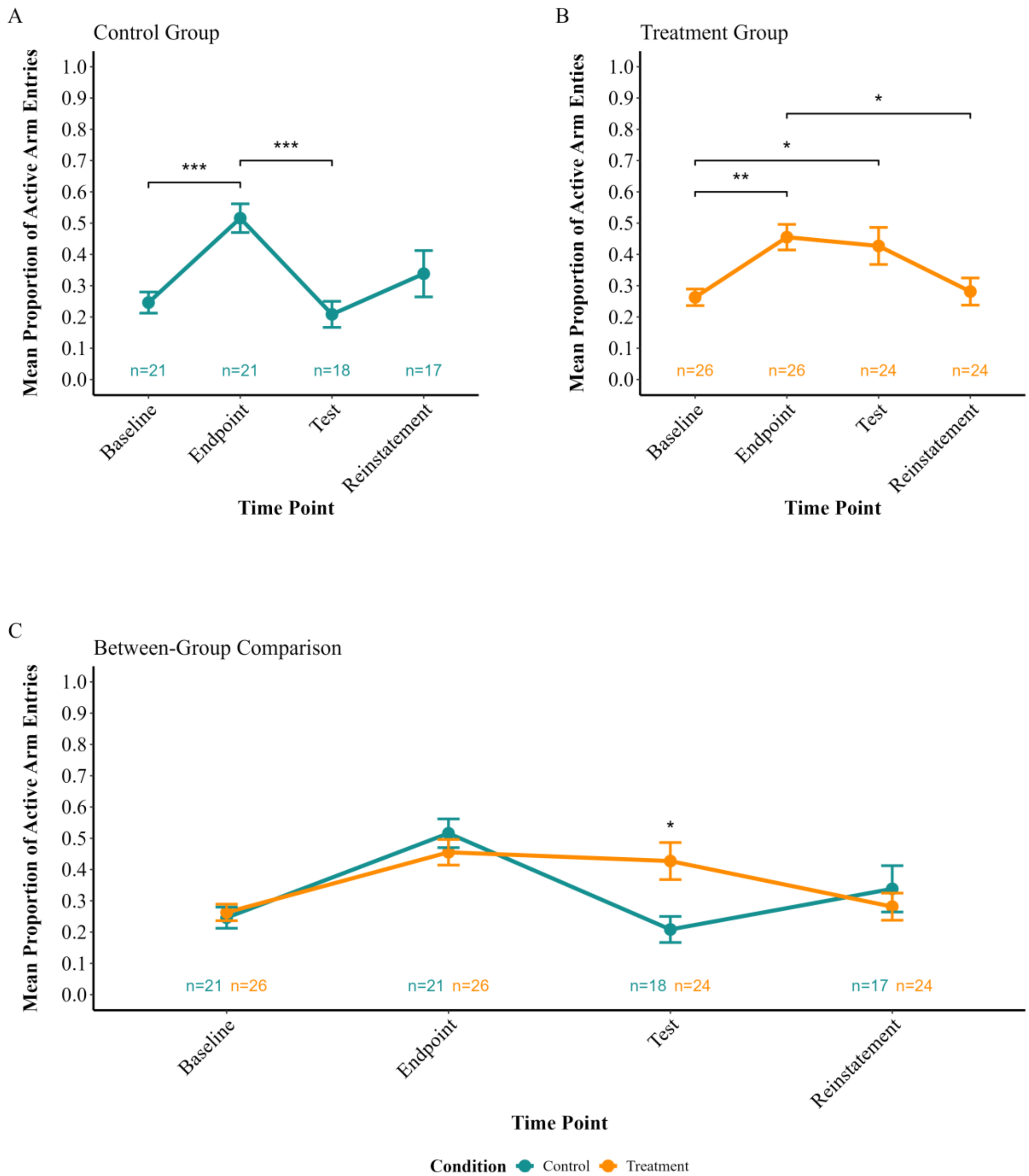
We did not detect a significant effect of condition (χ^2 (1) = 0.773, p = .379). The results indicated a significant effect of time (χ^2 (3) = 35.5, p < .001) and a significant time*condition interaction (χ^2 (3) = 10.2, p = .017).

Post-hoc pairwise comparisons were carried out using estimated marginal means with a Bonferroni corrections applied to account for multiple comparisons. Comparisons looked at within-group differences in response probability across the four phases and between-group differences at each phase. Note that, for the statistical comparisons, only the last 6 trials of the conditioning period were used (referred to as "endpoint"). The effect sizes were reported using Cohen's h which is appropriate when comparing two proportions (Cohen, 1988). There were two within-group differences for the control subjects: endpoint differed significantly from baseline (h

= 0.56, $p < .001$), and test differed significantly from endpoint ($h = 0.65$, $p < .001$). These differences represent medium effect sizes (Cohen, 1988, pp. 184–185). There were three within-group differences for the treatment subjects: endpoint differed significantly from baseline ($h = 0.4$, $p = .002$), test differed significantly from baseline ($h = 0.35$, $p = .044$), and reinstatement differed significantly from endpoint ($h = 0.36$, $p = .035$). These differences represent small effect sizes, with the baseline-to-endpoint difference approaching the medium effect size criterion of $h = 0.5$ (Cohen, 1988, pp. 184–185). A significant between-group difference was found in the preference score between treatment and control subjects at test ($h = 0.48$, $p = .004$) which represents a small to medium effect size. Treatment subjects entered the active arm more often ($M = 0.427$, $SD = 0.29$) than control subjects ($M = 0.208$, $SD = 0.177$). No other significant between-group differences were detected.

Figure 9

Learning and Memory Retention in Cocaine Treated Planaria vs Controls



Note. Changes in Y-maze active arm entries across experimental phases. The baseline phase included 6 trials, endpoint included the final 6 trials of conditioning, and both test and reinstatement phases included 4 trials each. A) The control group showed significant differences in active arm entries between baseline and endpoint, and between endpoint and test. B) The treatment group demonstrated significant differences between baseline and endpoint, baseline and test, and endpoint and reinstatement. C) Between-group comparisons revealed there was a significant difference in active arm entries at the test phase. Error bars represent standard error of the mean. * = $p < .05$; ** = $p < .01$; *** = $p < .001$.

Looking at the results in Figure 9B, subjects in the treatment group showed some evidence of a change in behaviour. These subjects were more likely to choose the active arm at the end of conditioning (endpoint) compared to baseline. Moreover, this preference was maintained for two weeks as demonstrated by the heightened active arm entries at test. Despite the behavioural change persisting for two weeks, when tested the next day during the reinstatement procedure the proportion of active arm entries had returned to baseline levels. When considering these results we must note that, during conditioning trials, subjects would often move into the divot at the end of the runway and swim in circles. The circling behaviour could have impacted the sense of direction for subjects. Reinforcement provided on these trials may have been ineffective and could have limited the extent of learning among treatment subjects.

The control group demonstrated a similar increase in active arm entries despite receiving no reinforcement (Figure 9A). However, in contrast to the treatment group, this returned to baseline levels when tested two weeks after conditioning. The behaviour of control subjects highlights the natural variability in planaria behaviour over time.

The between-groups comparison seen in Figure 9C shows a significant difference between groups at test. The stable change in behaviour shown by the treatment group may be evidence of true learning in contrast to the natural variability of behaviour seen in the control group. This could explain why the change in behaviour for the treatment group persisted for two weeks, while the behaviour of the control group diminished back to baseline levels. Admittedly, the reinstatement results reduce the credibility of this explanation. If the conditioned behaviour was able to persist for several weeks, it is puzzling that it would be extinguished so rapidly yet fail to be reinstated. Overall, the results provide preliminary support for the notion that planaria can be conditioned in a Y-maze and that this response can last at least two weeks. There was no evidence that learning can be reinstated by re-exposure to the reinforcing compound after regeneration.

This experiment also considered response times for each trial. We hypothesised that, even if planaria cannot learn to navigate to the reinforced arm, they may demonstrate increased motivation if the maze environment serves as a cue that a reward is available. This could be inferred from faster responding in the treatment group compared to the control group. The response time data across the four experimental phases are shown in Figure 10.

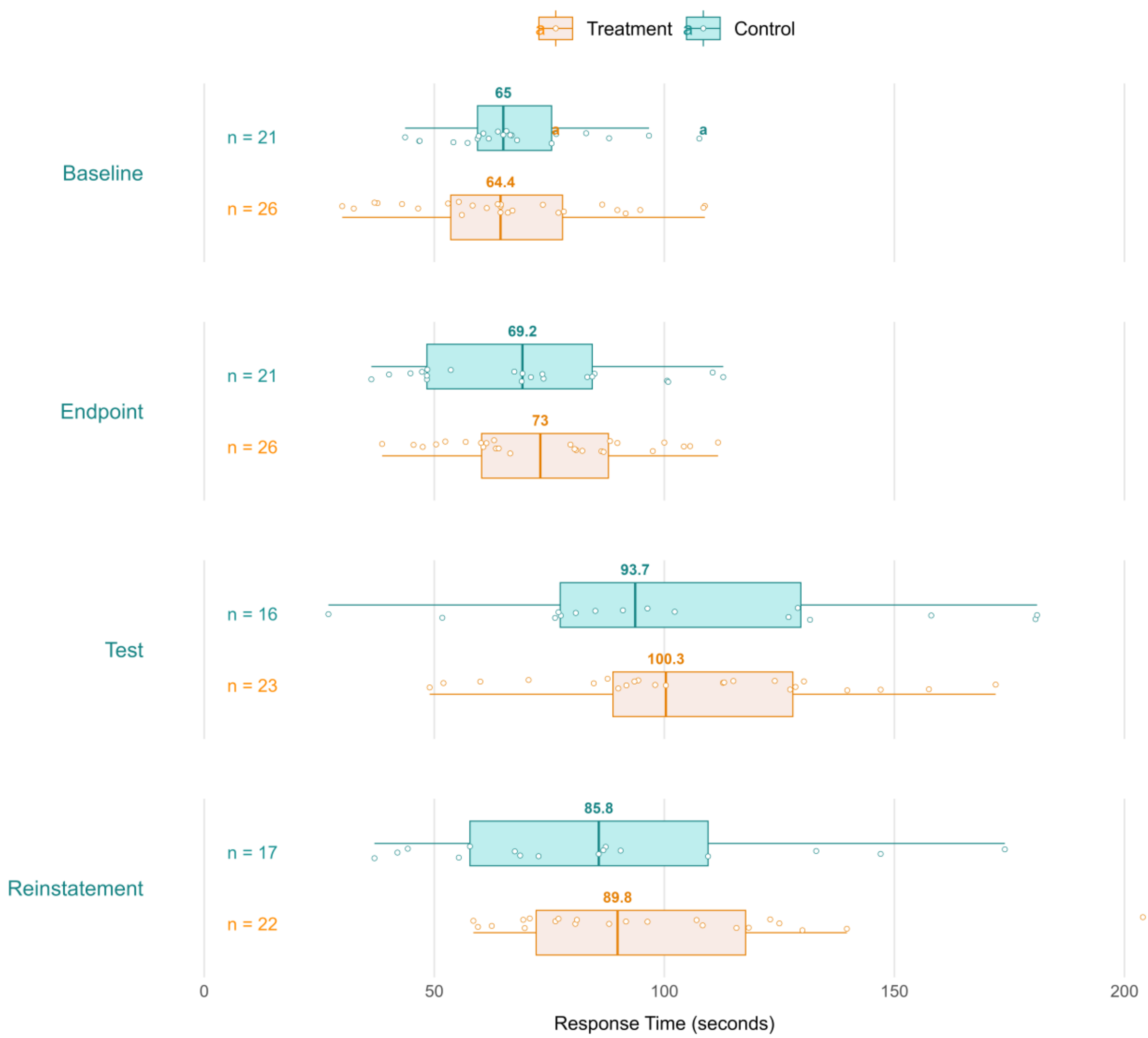
The decision latency data were analysed with a linear mixed-effects model using the lme4 package for R (Bates et al., 2015). The model included fixed effects of condition, time point, and an interaction term. Subject ID was set as a random effect to account for repeated measures. The decision time data were log-transformed. Type III ANOVA was conducted using the car package (Fox & Weisberg, 2019) to test for statistical significance of the fixed effects. Pairwise comparisons with a Bonferroni correction were carried out using the emmeans package in R (Lenth, 2024).

The ANOVA results revealed a significant effect of time (χ^2 (3) = 24, p < .001) but failed to show a significant effect of condition (χ^2 (1) = 0.266, p = .606). No time*condition interaction effect was found (χ^2 (3) = 1.88, p = .598).

Post-hoc pairwise comparisons were carried out using estimated marginal means with Bonferroni corrections applied to account for multiple comparisons. Comparisons looked at within-group differences in decision latency across the four phases and also looked at between-group differences at each phase. The results indicated that there were two within-group differences for the control subjects: test differed significantly from baseline (d = .56, p < .001) and test differed significantly from endpoint (d = .54, p < .001). There were four within-group differences for the treatment subjects: test differed significantly from baseline (d = .85, p < .001), reinstatement differed significantly from baseline (d = .67, p < .001), test differed significantly from endpoint (d = .53, p < .001), and reinstatement differed significantly from endpoint (d = .40, p = .007). There were no significant differences between groups.

Figure 10

Decision Latency Across Experimental Phases



Note. Time taken to make a response across phases and between conditions. Vertical lines show the median values. Boxes represent the inter-quartile range (IQR), with whiskers extending out 1.5 times the IQR.

The decision latency data do not support our hypothesis. Instead of decision latency decreasing for the treatment group between baseline and the end of conditioning as predicted, there was no change for either treatment or control groups. Moreover, both groups showed increased latency for decision making when comparing test to baseline. To understand these results, it is important to consider that the experimenter observed that planaria became smaller during the experiment. This has been reported to occur when planaria have been deprived of food for prolonged periods. It is thought to be an evolved survival mechanism in planaria (Pascual-Carreras et al., 2020; Thommen et al., 2019). Interestingly, some research suggests that planarian locomotion is not correlated with body size (Raffa et al., 2001). But this may only hold for between-subjects comparisons during short experiments but not within-subject comparisons.

The change in body size that was observed throughout the experiment is likely a sign of impaired health and low energy availability. One cause of poor health may have been the repeated handling with a paintbrush, which was suspected to have resulted in the death of several subjects. While the remaining subjects could still complete the maze, their movement may have been impaired due to injuries. This may have obscured any effect of increased motivation.

These decision latency results are compatible with at least three interpretations. One possibility is that there was no increase in motivation among planaria. This hinders the claim that the change in behaviour for treatment subjects was evidence of learning. Another possibility is that response latency is a useful measure of motivation, but the effect of fatigue and injury in this case impaired the ability to detect increased motivation. Finally, response latency may simply not be a suitable measure of planarian motivation. Given the time and effort required to record response latency despite not being sure of its validity for measuring motivation, it was not measured during subsequent experiments.

Experiment 3

The Y-maze experiment carried out in Experiment 2 suffered from several procedural issues. First, the Y-maze contained a divot at the end of the runway that planaria would often swim into and circle around – possibly disturbing any sense of direction relative to the start of the maze. Second, due to handling planaria with a paint brush, the death rate was relatively high. Third, the number of trials where subjects failed to respond was high, perhaps due to fatigue from too many trials per day. The effect of fatigue on responses in planaria has been discussed in the literature (Best & Rubinstein, 1962; Lee, 1963). Finally, the behavioural change was relatively modest in size. To convincingly test memory retention, it would be preferable to establish a stronger and more stable memory. Despite these limitations, there was some indication that planaria were able to learn and retain a conditioned response and that this may endure for at least two weeks. Because of this, we decided that the project would progress to investigating retention through regeneration.

A third experiment was thus carried out which aimed to improve on the limitations described for experiment 2 and test whether memories can be retained through regeneration. Because of the modest behaviour change seen in Experiment 2 when cocaine was used, and due to successful learning achieved by another colleague using methamphetamine to condition planaria (see Ramirez, 2025), this project adopted methamphetamine as the reinforcing stimulus for the remaining experiments. Methamphetamine is a psychoactive compound similar to cocaine in that it also increases extracellular dopamine levels (Desai et al., 2010; Kuczenski et al., 1995). Methamphetamine has been established elsewhere as a rewarding stimulus for planaria in models of addiction and withdrawal (Kusayama & Watanabe, 2000; Raffa et al., 2008; Sacavage et al., 2008). Similar to the results presented in Experiment 1, Ramirez (2025) demonstrated that methamphetamine concentrations between 1 µM and 20 µM did not significantly affect motility.

Colony Maintenance and Handling

The planaria and colony maintenance protocols were similar to those described in Experiment 2. The key differences were that the colony was stored in a small plastic container, and once experimental subjects had been collected they were stored in a separate experimental room. This room was dimly illuminated with red light during experimentation and otherwise kept dark. Planaria length was not measured, but only those visually estimated to be 1 cm long were selected for this experiment. Planaria were handled using a plastic transfer pipette with the tip cut off.

Materials and Procedure

Forty-two planaria were used for this experiment, all assigned to the methamphetamine treatment group. There was no control group[5]. This experiment had four experimental stages: baseline, conditioning, retention test, and reinstatement (see Figure 11). Planaria completed three trials per day. There were two days for baseline, five days for conditioning, and one day each for the retention test and reinstatement procedure. During baseline and conditioning trials multiple planaria were run concurrently in separate Y-mazes.

Each maze was filled with 2 ml of planaria water which was shaken gently to evenly distribute the water throughout the runway and arms. Planaria completed three trials per day with an intertrial interval of approximately 120 minutes. At the start of a trial, planaria were transferred to the start of the maze runway using a plastic transfer pipette with the tip cut off. A timer was started once each planarian was placed in its maze runway. Planaria were given up to five minutes to enter one of the arms. Once a planarian had entered an arm, the plug was inserted to stop liquid moving between compartments, after which approximately 0.675 ml remained in each arm[6]. A planarian was considered to have entered the arm when the plug could be safely inserted without touching the planarian.

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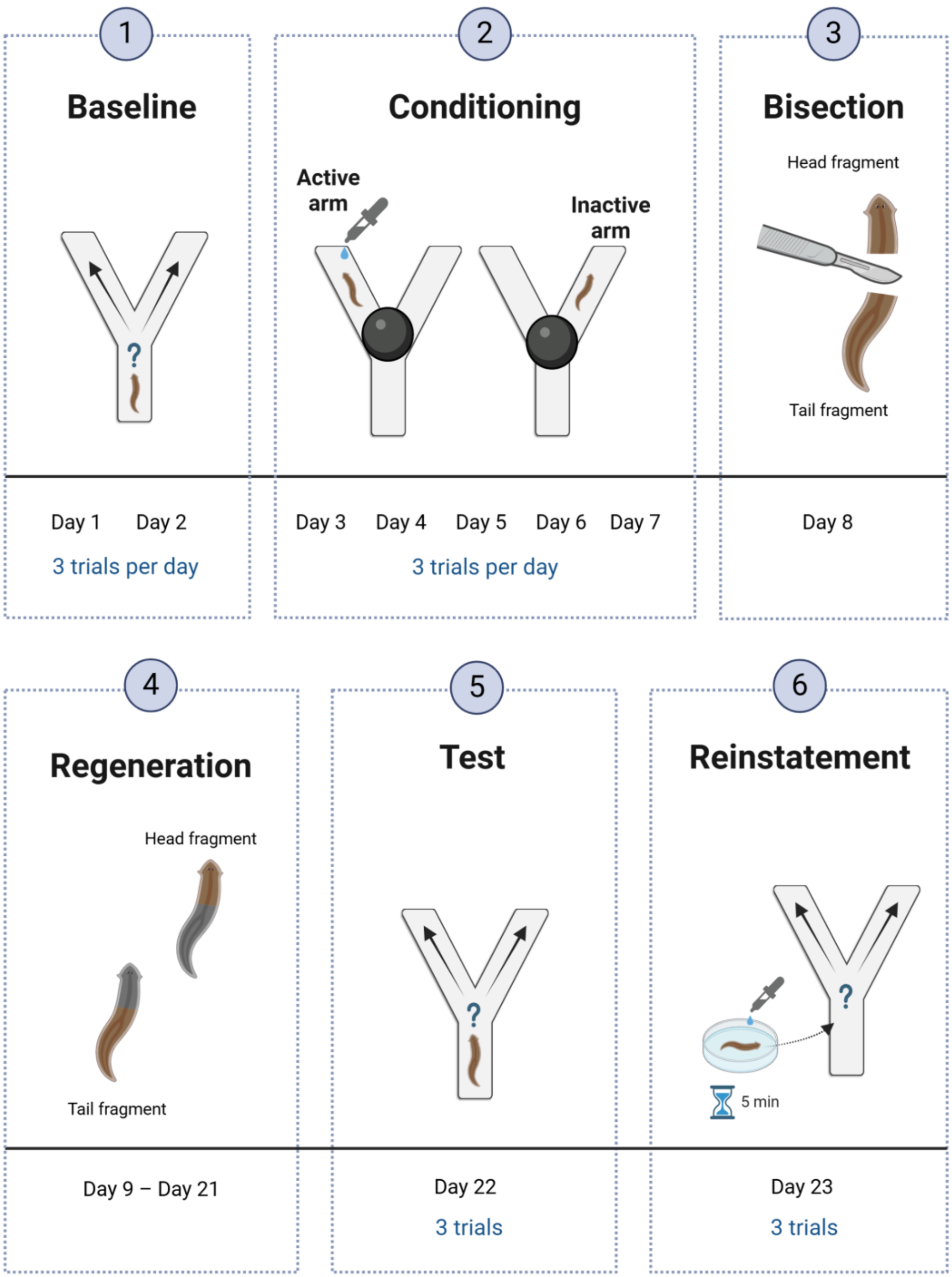
[5] It was important to establish whether an increase in active arm entries is stable over the conditioning period. To allow for a large number of treatment subjects to be included, no control group was run

[6] Measured by extracting and weighing the volume of distilled water trapped in each arm, assuming a density of 1 g/ml. Small deviations are inevitable, but this was the most common value seen during our tests

Figure 11

Graphical Timeline of Experiment 3



When treatment subjects entered the active arm, 29.35 µL of methamphetamine (BDG Synthesis, Wellington, New Zealand) in distilled water was pipetted throughout the arm to achieve a 10 µM concentration. Nothing was added when planaria entered the inactive arm. After an arm was chosen and the drug or vehicle was administered, the timer was restarted and three minutes were given for absorption. The runway light was turned on after placing the subject in the maze and turned off once the subject entered an arm. At the end of each trial, planaria were washed thoroughly in a bath of planaria water before being transferred back to their 12-well dish. Mazes were rinsed and dried between each trial.

Planaria which showed evidence of learning at the end of conditioning were bisected on the day following their last conditioning trial (n = 10). Evidence of learning was assessed as having entered the active arm during at least 4 of the last 6 conditioning trials. For the bisection, planaria were placed individually onto a plastic block with some planaria water and cut transversely using a flat edge blade. The cut was made above the pharynx and below the base of the head (see Figure 13). This resulted in two subjects: a tailless head fragment and a headless tail fragment. Planaria fragments were placed into 12-well plates with labels to track the original subject number and whether they were a head or tail fragment. Bisected subjects were not fed for the duration of the experiment.

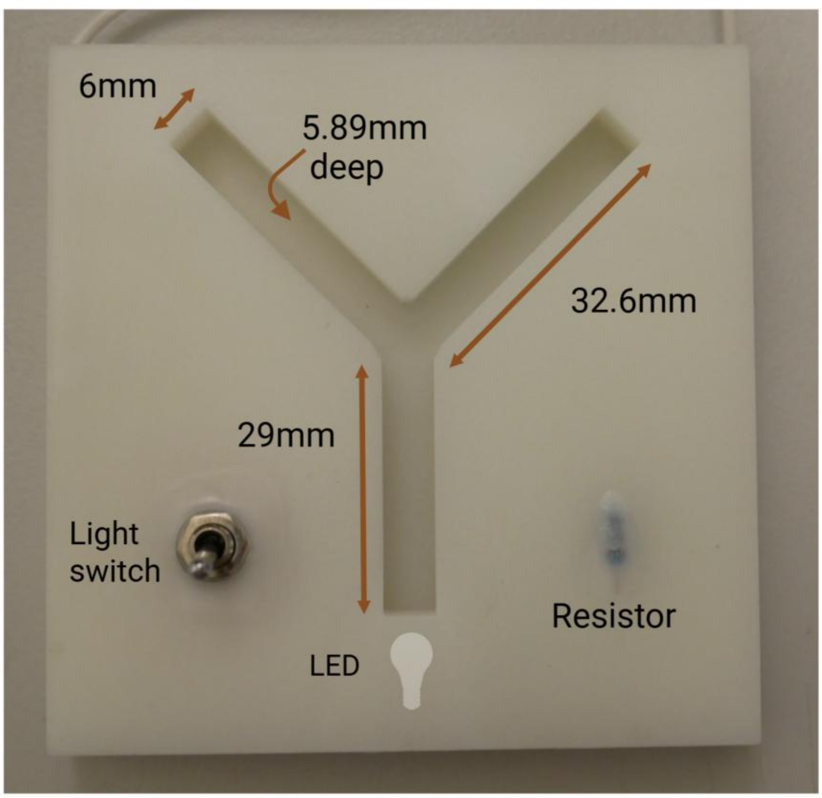
To enable sufficient time for regeneration of the head and tail fragments, the memory retention test took place 14 days after bisection (see Figure 13 for an overview of the regeneration timeline). For the memory retention test on day 14 of regeneration, multiple planaria were run simultaneously in separate Y-mazes. Planaria were given five minutes to enter an arm. Once a decision was made, the plug was inserted and planaria were left for approximately 60 seconds before being moved back to the holding dish. No additional liquid was added during test trials. The inter-trial interval was approximately 30 minutes. A reinstatement procedure was carried out the following day. The reinstatement procedure was identical to the test stage with the only additional step being drug exposure before each trial. At the start of a run, planaria were placed into separate 4 ml solutions of planaria water containing 10 µM of methamphetamine for five minutes. At the end of the exposure interval, each planarian was moved into a Y-maze to begin the first trial.

Four Y-mazes were 3D printed for this experiment (for dimensions see Figure 12). Mazes were printed using Siraya Tech professional UV resin. Similar to the mazes used in Experiment 2, these mazes also contained an LED light embedded in the resin after printing to induce negative phototaxis.

As described in Experiment 2, three exclusion criteria were used. The exclusion criteria were: A) failing to complete at least four of the six baseline trials; B) failing to complete at least two trials on consecutive conditioning days; C) failing to complete at least four of the last six trials of conditioning. None of the subjects were excluded based on those criteria. No subjects died during the experiment.

Figure 12

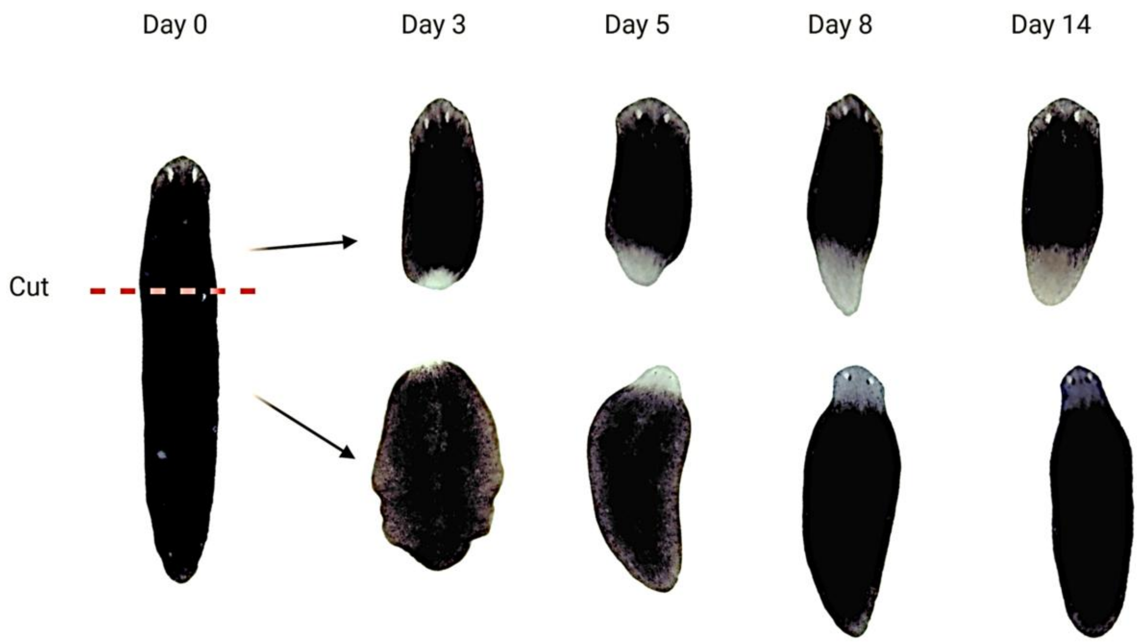
Modified 3D Printed Y-maze



Note. Four Y-mazes (one pictured above) were printed using Siraya Tech professional UV resin. A white LED was embedded in the resin at the start of the runway to induce negative phototaxis. The light was powered by a single 9V battery. A silicone plug (not shown) was moulded to block liquid transfer after a planarian entered one of the arms.

Figure 13

Planaria Regeneration Timeline



Note. The image above shows the regeneration process for one of the experimental subjects used in this project. Soon after bisection, a blastema containing stem cells appears at the wound site which gives it the white colouring. The contrast and sharpness of the figure have been adjusted to make the regenerating blastema more visible. The regeneration process has been described in detail by Reddien and Alvarado (2004).

Results and Discussion

Most subjects had an initial preference towards the right arm (n = 22), with just over a quarter favouring the left arm (n = 13) and a few having no preference (n = 7). This experiment employed a biased design so that the active arm to be reinforced was the opposite of the initial preference or randomly assigned for those with no initial preference. The left arm was active for 25 subjects, and the right arm was active for 17 subjects.

Figure 14A shows the change in active arm preference across conditioning days for all subjects ($n = 42$). An increase in active arm entries was visible on the first day of conditioning. This change persisted over the remaining days, although there was a slight downward trend as conditioning proceeded. A paired t-test was used to test whether there was a significant difference in the proportion of active arm entries between baseline and the end of conditioning. For the statistical comparison, only the last 6 trials of the conditioning period were included (referred to as "endpoint"). At the end of conditioning, planaria entered the active arm significantly more often than before conditioning ($t(41) = 4.53$, $d = 0.7$, $p < .001$). This change in preference between baseline and endpoint can be seen in Figure 14B.

Figure 14C and D show the memory retention and reinstatement data for bisected planaria ($n = 10$) that were given 14 days to regenerate. Paired t-tests were used to assess whether the active arm preference after regeneration and during reinstatement were significantly different from baseline. For the post-regeneration memory test, we failed to find a difference between the heads ($t(9) = 1.72$, $d = 0.54$, $p = .12$) or tails ($t(9) = 1.86$, $d = 0.59$, $p = .096$) compared to the baseline active arm entries. Similarly, the reinstatement results failed to demonstrate a significant difference between the heads ($t(9) = 1.34$, $d = 0.42$, $p = .215$) or tails ($t(9) = 0.67$, $d = 0.21$, $p = .521$) compared to the baseline active arm entries.

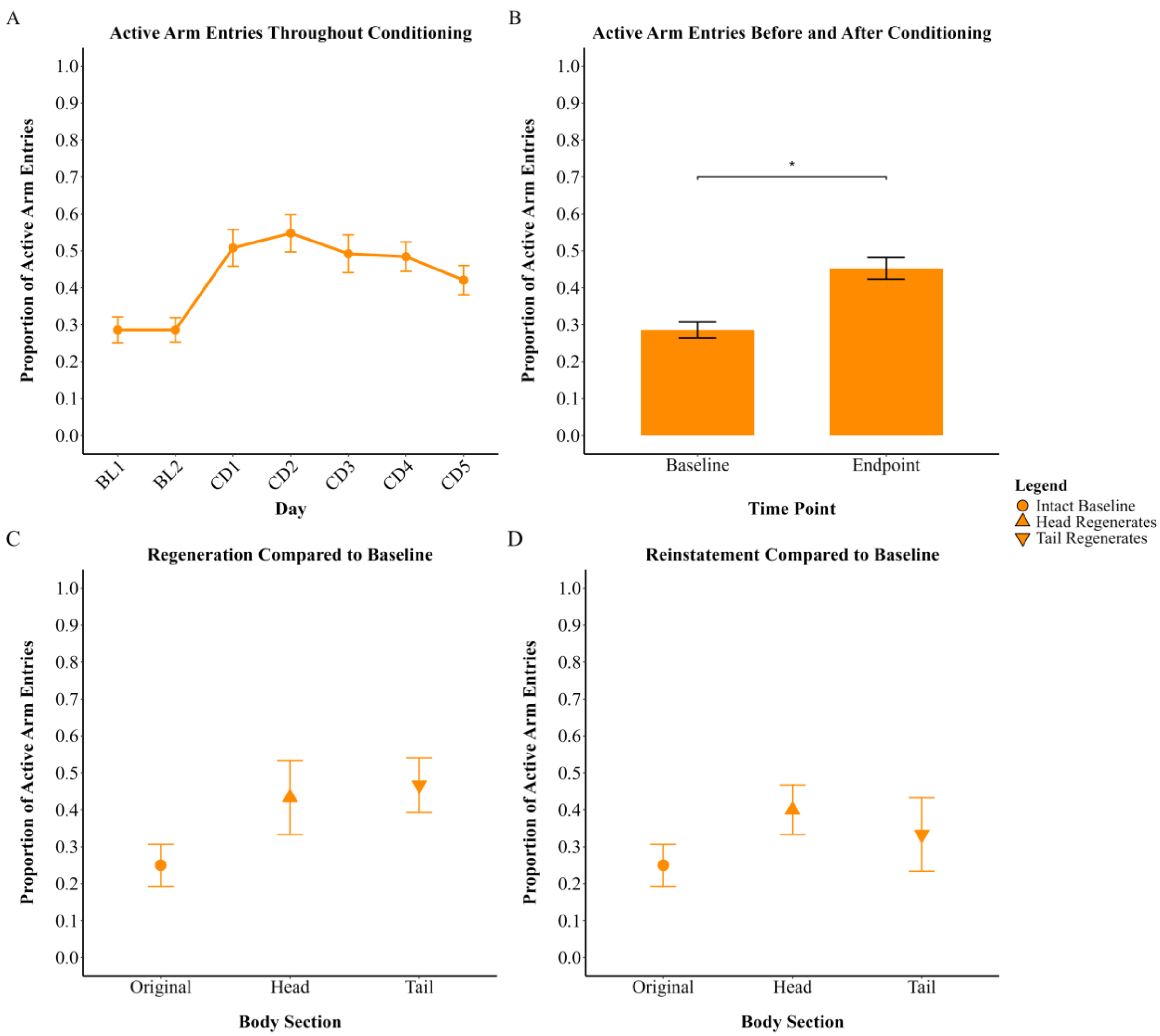
The results described here provide preliminary evidence that methamphetamine can be used to shape the behaviour of planaria in a Y-maze. The behavioural change took place rapidly, with an increase in active arm entries visible after one day of conditioning. The speed at which behaviour change takes place is likely conditional on when planaria sample the active arm. By chance, many subjects in one group may enter the active arm on the first or second trial whereas in subsequent groups it may take subjects three or four trials. While a gradual increase in the active arm entries was expected, this rapid increase can be explained by the fact that 22 subjects entered the active arm on the first trial.

It could be argued that what appears to be a change in preference towards the active arm (i.e., successful conditioning) is actually the expression of truly random behaviour. Given a binary choice, if the planaria were behaving randomly you would expect them to enter each arm 50% of the time on average. Because the proportion of active arm entries centred around 0.5 during conditioning, this could reasonably explain the observed behaviour. In contrast, if the behaviour was random, then we must explain why the baseline level of active arm entries was low across two consecutive conditioning days. If planaria do not typically have a directional preference, we should have seen more variation between day one and two during baseline. Moreover, as the frequency of active arm entries did not return to baseline levels at all throughout conditioning, we cannot easily attribute this pattern to chance. As no control group was run, it is not clear whether a similar pattern would occur without active reinforcement in these mazes.

Although neither the head nor tail regenerates differed significantly in their active arm entries compared to baseline, visual inspection showed that both group means appeared higher than baseline (see Figure 14C and D). For example, the proportion of active arm entries for tail regenerates (*M* = 0.467, *SD* = 0.233) was similar to what was seen at the end of conditioning for the original sample (*M* = 0.452, *SD* = 0.189). It must of course be noted that the failure to find a significant effect despite the apparent trends may be due to the small sample size for regenerates (*n* = 10). Larger samples in the future may help identify subtle effects if they in fact exist. Nevertheless, the data indicated that, although planaria can be conditioned using a Y-maze, there was no statistical evidence that this learned response survived bisection and regeneration. A stronger test of the hypothesis would require both a control group and larger sample sizes. Moreover, it may be beneficial to use a higher dose in future experiments as prior research suggests that methamphetamine can increase the strength of learning in a dose dependent manner (Kusayama & Watanabe, 2000).

Figure 14

Learning and Memory Retention in Methamphetamine Treated Planaria



Note. The baseline phase included 6 trials, the endpoint included the final 6 trials of conditioning, and both test and reinstatement phases included 3 trials each. A) shows the mean proportion of active arm entries for each day during baseline and conditioning. B) Methamphetamine treated subjects showed a significant difference in their active arm entries between baseline and endpoint. C) Despite a visual trend showing greater active arm entries for regenerates compared to baseline, these differences were not statistically significant. D) The reinstatement procedure failed to generate significant differences in active arm entries compared to baseline. Error bars represent standard error of the mean. BL = baseline; CD = conditioning; * = p < .05.

Experiment 4

The Y-maze experiment carried out in Experiment 3 made several improvements on the materials and procedure from Experiment 2. Yet the results were difficult to interpret due to a number of limitations. Specifically, it did not employ a control group, a small number of subjects were used for the regeneration tests, and the change in active arm entries was significant but still relatively weak. To address these limitations, this experiment included a control group, used more subjects in the regeneration phase, and used a higher dose of methamphetamine for reinforcement.

Colony Maintenance and Handling

The planaria maintenance and handling protocols were identical to those described in Experiment 3. Planaria length was not measured, but only those visually estimated to be 1 cm long were selected for this experiment.

Materials and Methods

This experiment used two groups: a treatment group (n = 15) which received methamphetamine and a control group (n = 15) which received vehicle only. There were four experimental stages: baseline, conditioning, regeneration test, and reinstatement (see Figure 11). The materials and procedures used here were the same as Experiment 3 except for the following modifications. The active arm was reinforced with 58.7 µL of methamphetamine (BDG Synthesis, Wellington, New Zealand) in distilled water to reach a final concentration of 20 µM. Control subjects received an equal volume of vehicle only (distilled water) when they entered the active arm, and were otherwise treated the same as the treatment subjects. At the end of conditioning, all subjects were bisected into head and tail halves. For conditioning and testing procedures, the intertrial intervals were approximately 90 minutes and 60 minutes, respectively. For the reinstatement phase, all regenerated planaria, including control subjects, were bathed in 20 µM methamphetamine prior to each trial. The exclusion criteria employed were the same as in Experiment 2. None of the subjects used for this experiment were excluded based on those criteria. No subjects died during this experiment.

Results and Discussion

There was a slight bias in baseline arm preference with 12 subjects favouring the right arm, 9 favouring the left arm, and 9 having no preference. This experiment employed a biased design such that the active arm to be reinforced was the opposite of the initial preference or randomly assigned for those with no initial preference. The left arm was active for 15 subjects, and the right arm was active for 15 subjects.

Figure 15A shows the proportion of active arm entries throughout conditioning for both treatment (n = 15) and control (n = 15) subjects. The treatment group exhibits a steady increase in active arm entries over the first three conditioning days. However, this begins to decrease sharply after day three. The control group shows little variation across conditioning.

To assess whether the active arm preference at the end of conditioning (endpoint) differed from baseline, a generalised linear mixed effects model with family set to binomial was fitted in R using the lme4 package (Bates et al., 2015). Subject ID was set as a random effect, with condition, time point and the interaction term as fixed effects. The model analysed the proportion of entries into the active arm out of six total trials at each time point. Pairwise comparisons with a Bonferroni correction were carried out using the emmeans package in R (Lenth, 2024). A Type III ANOVA was conducted using the car package (Fox & Weisberg, 2019) to identify whether there was a significant effect of condition or time, or an interaction effect.

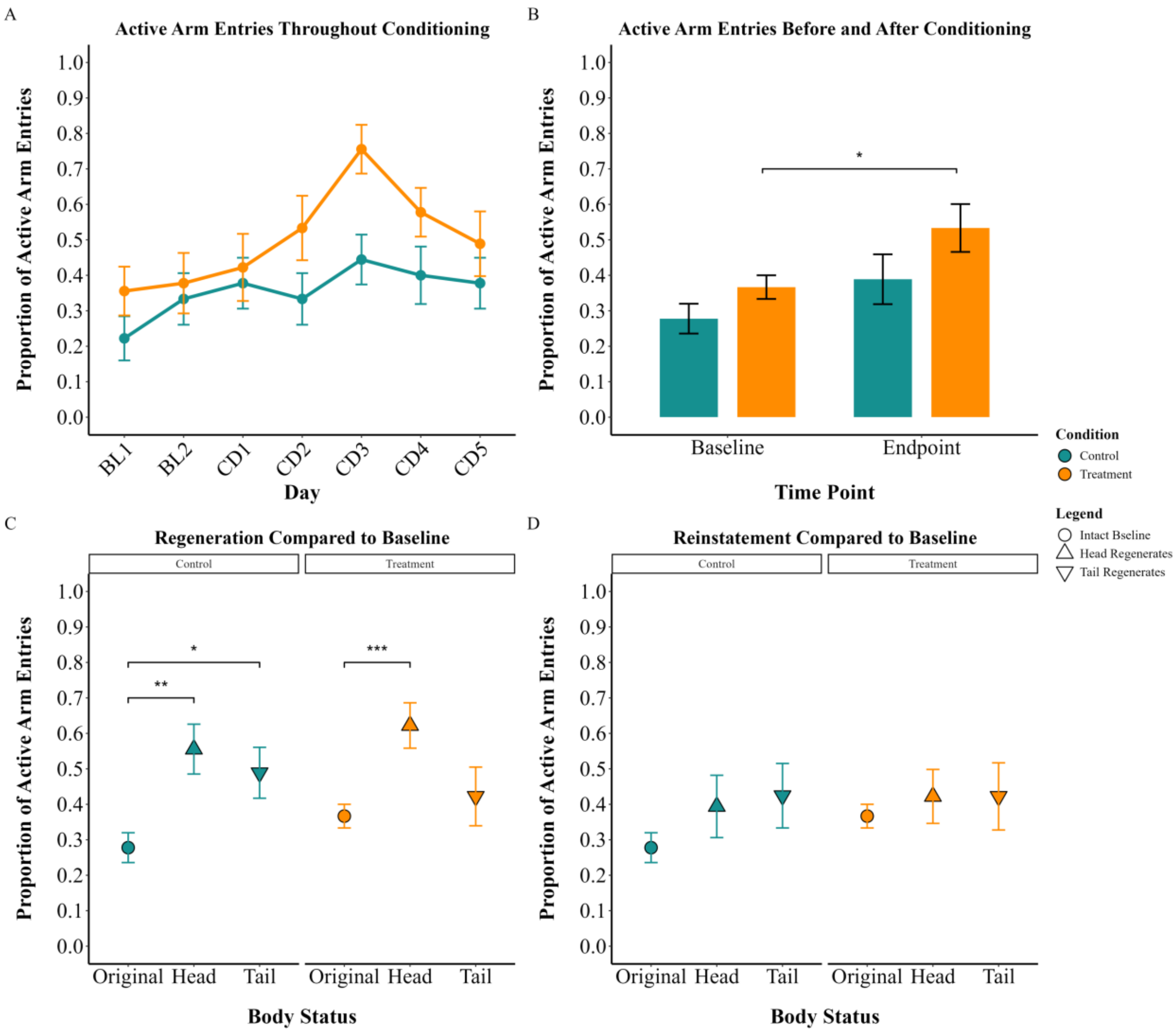
There was a significant main effect of time ($\chi^2(1) = 7.179$, $p = .007$). and a significant main effect of condition ($\chi^2(1) = 5.021$, $p = .025$). We did not detect a significant condition * time interaction ($\chi^2(1) = 0.16$, $p = .689$). Post-hoc pairwise comparisons were carried out using estimated marginal means with a Bonferroni corrections applied to account for multiple comparisons. Comparisons looked at within-group differences in response probability over time and between-group differences at each time point. The within-group comparison showed that endpoint was significantly higher compared to baseline for treatment subjects only ($OR = 0.51$, $z = 2.23$, $p = .025$). The post-hoc comparisons failed to show any significant between-group differences at baseline or endpoint. This indicates that while active arm entries were higher for the treatment group overall, the individual data points were not significantly different when comparing the treatment group to the control group.

Paired t-tests were used to determine whether there were significant within-group differences when comparing memory retention and reinstatement data to baseline active arm entries (see Figure 15 C and D). For the initial memory retention test on day 14 the control heads differed significantly from their baseline active arm preference ($t(14) = 3.31$, $d = 0.85$, $p = .005$). Treatment heads differed significantly from their baseline active arm preference ($t(14) = 4.07$, $d = 1.05$, $p < .001$). Control tails differed significantly from their baseline active arm preference ($t(14) = 2.24$, $d = 0.34$, $p = .042$). For reinstatement on day 15, no significant differences were found between heads or tails and their respective baseline values for either group.

The data presented in Figure 15 support the claim that methamphetamine can be used to successfully shape the responses of planaria in a Y-maze. In particular, Figure 15A shows a gradual increase in active arm selection for the treatment group across the first three days of conditioning. Similar to the trend seen in Experiment 3 (Figure 14A), treatment subjects exhibited a decrease in active arm entries towards the end of conditioning. The control group showed little change in their behaviour during conditioning.

Figure 15

Learning and Memory Retention Through Decapitation in Planaria



Note. The baseline phase included 6 trials, endpoint included the final 6 trials of conditioning, and both test and reinstatement phases included 3 trials each. A) The mean proportion of active arm entries each day during baseline and conditioning B) Methamphetamine treated subjects showed a significant difference in their active arm entries between baseline and endpoint. C) The day 14 retention test found that control head and tail regenerates, and treatment head regenerates differed significantly from baseline. D) No significant differences were observed between regenerates of either group and their respective baseline values during reinstatement. Error bars represent standard error of the mean. BL = Baseline; CD = Conditioning; * = $p < .05$; ** = $p < .01$; *** = $p < .001$.

Experiment 5

The experiments described above support the claim that planaria can obtain an operantly conditioned response. Yet, as can be seen from Figure 9, Figure 14 and Figure 15, the response was not stable. The conditioned response tended to reach its peak early during conditioning and then diminished towards baseline values despite still being actively reinforced. It is not clear whether the reversion to baseline behaviour is a sign of forgetting, drug tolerance, or active rejection of a reinforced direction. Active rejection being the dramatic return to baseline behaviour or below after learning has occurred, despite continued reinforcement for the desired behaviour (described in Best & Rubinstein, 1962, pp. 565–566). To properly test memory retention through regeneration, it is important to have a high rate of successful responses just prior to bisection. Otherwise, what appears to be a lack of retention in regenerates may just be a continuation of an already declining response rate. An additional Y-maze experiment was therefore carried out.

Rather than having a pre-determined conditioning period, it was decided that data would be inspected each day for evidence of learning. A mean proportion of active arm entries above 0.6 for the treatment group would trigger an end of the conditioning period. This was to ensure the active arm was preferred by planaria so as to give the greatest chance of retaining the learned response throughout the regeneration period. Initiation of the regeneration phase was thus conditional on whether adequate learning was shown for the treatment subjects.

Colony Maintenance and Handling

The planaria maintenance and handling protocols were identical to those described in Experiment 3. Planaria length was not measured, but only those visually estimated to be 1 cm long were selected for this experiment.

Materials and Procedure

This experiment used two groups: a methamphetamine treated group (n = 24) and a control group (n = 24) which received vehicle only. This experiment had two stages: baseline and conditioning[7]. The materials and procedures used here for baseline and conditioning were identical to those in Experiment 4. The exclusion criteria employed were the same as in Experiment 2. None of the subjects used for this experiment were excluded based on those criteria. No subjects died during the experiment.

The mean proportion of active arm entries for treatment and control subjects were monitored each day. The data were graphed to show daily active arm entries compared to baseline. No statistical analyses were performed during the conditioning period. After day four of conditioning the experiment was stopped due to a consistent decrease in active arm entries among the treatment group.

Results and Discussion

An approximately equal number of subjects preferred the right arm (n = 20) and left arm (n = 21), with a small number of subjects having no preference (n = 7) at baseline. This experiment employed a biased design, such that the active arm to be reinforced was the opposite of the initial preference, or randomly assigned for those with no initial preference. The left arm was active for 23 subjects, and the right arm was active for 25 subjects.

Figure 16A shows the change in active arm preference across conditioning days for both treatment (n = 24) and control (n = 24) subjects. The treatment group demonstrates a sudden jump on the first day of conditioning. This change remains stable for one day before declining towards baseline levels. The control group shows a steady increase across days, with a slight decrease on the last day of conditioning. For statistical comparisons, only the last 6 trials of the conditioning period were included (referred to as “endpoint”).

To assess whether the endpoint active arm preference differed from baseline, a generalised linear mixed effects model with family set to binomial was fitted in R using the lme4 package (Bates et al., 2015). Subject ID was set as a random effect, with condition, time point and the interaction term as fixed effects. The model analysed the proportion of entries into the active arm out of six total trials at each time point. Pairwise comparisons with a Bonferroni

[7] The treatment subjects failed to show adequate evidence of learning. Given the failure of learning and the extended time required to perform the regeneration phase, it was decided that the experiment would be terminated after day four of conditioning.

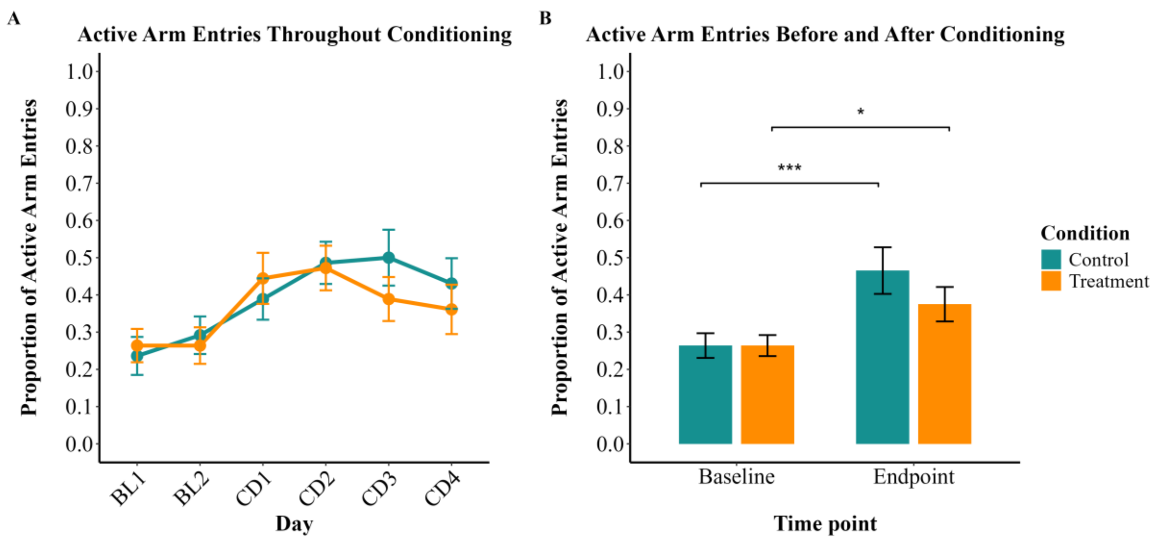
correction were carried out using the emmeans package in R (Lenth, 2024). A Type III ANOVA was conducted using the car package (Fox & Weisberg, 2019) to identify whether there was a significant effect of condition or time, or an interaction effect.

There was a significant main effect of time ($\chi^2(1) = 15.511$, $p < .001$), but no significant effect of condition or a time * condition interaction. Post-hoc pairwise comparisons compared the proportion of arm entries at baseline to endpoint. After four days of conditioning with 20 µM of methamphetamine, treatment subjects were significantly more likely to enter the active arm ($OR = 0.59$, $z = 2.04$, $p = .042$) compared to baseline. After four days of conditioning with vehicle only, control subjects were significantly more likely to enter the active arm ($OR = 0.4$, $z = 3.55$, $p < .001$) compared to baseline. We did not detect any significant between-group differences.

The results presented here conflict with those presented in Experiment 4. As can be seen in Figure 15, the active arm preference for the control group did not shift significantly, suggesting that in Experiment 4, the change observed in the treatment group was due to reinforcement with methamphetamine. However, the results presented here show a large shift in the control groups behaviour despite being treated with vehicle only. Further still, the control group appears to have experienced a greater shift in their preference than the treatment group (albeit the between-groups comparison at endpoint was not statistically significant). What appeared to be effective conditioning as a result of methamphetamine administration in Experiment 4 may instead be the result of some other extraneous other variable. Or, as described earlier, it may indicate that planaria do not typically have a directional preference in the Y-maze.

Figure 16

Failure to Find Adequate Evidence of Learning in Planaria



Note. Changes in Y-maze active arm preference across experimental phases. The baseline phase included 6 trials; the endpoint included the final 6 trials of conditioning. A) shows the mean proportion of active arm entries for baseline and conditioning. B) Both methamphetamine treated subjects and control subjects treated with vehicle only showed a significant difference in their active arm entries between baseline and endpoint. Error bars represent standard error of the mean. BL = Baseline; CD = Conditioning; * = p < .05; *** = p < .001.

General Discussion

Review of Main Findings

Planaria have gained attraction as a model organism in several areas of science ranging from drug addiction to limb regeneration. But the most unique area of planaria research involves the study of memory retention through decapitation and head regeneration. Despite a pockmarked past in the 20th century, recent experiments suggest that, even after losing their brain, planaria can retain previously acquired associative memories which can then be acted upon once a new brain is regenerated (Samuel et al., 2021; Shomrat & Levin, 2013).

This phenomenon is extraordinary in and of itself, but whether it has implications for the kinds of memories that concern humans has not yet been shown. For example, recent experiments have focused on things like familiarity with a surface texture and other simple associative memories. There have been no clear tests assessing the persistence of complex forms of memories formed through operant conditioning. Operant conditioning requires that a subject learns how to manoeuvre its body through the environment to receive a reward, just as humans must navigate through the world to accomplish their goals (e.g., getting to work on time).

A number of factors may explain the lack of research concerning whether operantly conditioned responses survive decapitation and brain regeneration. For example, researchers may have tried and failed to achieve conditioning with operant procedures but not published their findings. Alternatively, it may be possible to achieve for some subjects but not others, just as some humans excel in intellectual activities while others struggle. Planaria too may exhibit high variability in their cognitive capacities. This would make it difficult to reliably induce a conditioned response.

Another possible explanation for this research gap is the demanding schedule operant conditioning imposes on experimenters. Classical conditioning methods such as CPP often require a brief set up but then have periods of idle time while subjects are exposed to a stimulus, with software allowing for automatic tracking of movement. Operant conditioning, on the other hand, typically requires continuous observation of subjects to ensure a reward is delivered reliably and in close proximity to performance of the desired behaviour. When taking into account the necessary sample size and number of observations required per subject, operant conditioning procedures require a large time commitment.

The present research aimed to plug this gap in the planarian learning and memory literature. Specifically, it aimed to identify whether the phenomena of memory retention through decapitation and regeneration in planaria could be extended to more complex forms of memory. Additionally, it investigated whether time-dependent forgetting can be reversed with a reinstatement procedure.

We first established that a 20 μM dose of cocaine could be used for subsequent experiments without impairing the movement of planaria. In Experiment 2, we then investigated whether a Y-maze can be used to induce a conditioned response and whether this can persist for at least two weeks – the approximate period required for regeneration of planaria. Although both cocaine treated and control subjects entered the active arm more frequently at the end of conditioning, only the cocaine treated group showed evidence of a persistent change in behaviour when tested two weeks later. During the reinstatement procedure the following day, not only did reinstatement fail to reinstate or increase memory strength among the cocaine treated group, but the behaviour of the cocaine treated group actually returned to baseline levels. This suggested that, despite persisting for two weeks, the response was rapidly extinguished over three trials during the memory retention test on day 14.

Our observation of rapid extinction is consistent with the findings of Amaning-Kwarteng et al. (2017) who observed extinction over three trials in a CPP procedure. Moreover, recent work in our lab suggested that conditioned responses in the Y-maze are extinguished quickly when not reinforced (unpublished data, Invertebrate Neuroscience Lab). Contrary to expectations, the time taken to make a decision did not improve for the treatment group.

Over a series of experiments we then tested whether an operantly conditioned response can survive bisection and regeneration. Despite a visual trend in Experiment 3 which appeared to show memory retention through regeneration, we failed to find evidence of significant memory retention in regenerates. These results were followed up in Experiment 4 with a control group and larger sample size. We found evidence of memory retention after regeneration in the head regenerates of treatment subjects. Surprisingly, the control group (treated with vehicle only) also showed evidence of a change in behaviour after regeneration. That is, despite showing no statistically significant shift in behaviour during conditioning, the regenerated heads and tails of control subjects showed a high proportion of active arm entries after regeneration.

Although Experiment 3 and 4 implied successful conditioning of drug treated subjects, Experiment 5 failed to demonstrate adequate evidence of learning in the treatment group. Furthermore, the control group in Experiment 5 showed a significant shift in their behaviour. This shift was comparable in size to that observed among methamphetamine treated subjects from Experiments 3 and 4. It is apparent that each of the experiments described here, if inspected in isolation, would tell a different story.

Operant Conditioning in Planaria

Having briefly summarised the findings at the level of each experiment, we will now move to a general discussion of whether planaria can learn an operantly conditioned response in a Y-maze. It is tempting to take the observed shift in active arm entries for treatment subjects as evidence that we can successfully shape planarian behaviour. However, in Experiment 2 and 5 we also found a significant shift in the active arm entries of control subjects. A change in the behaviour of drug exposed subjects can be interpreted as successful reinforcement learning. But a change in the behaviour of control subjects is more difficult to understand, especially when it resembles a smooth learning curve.

It is possible that when control subjects received distilled water in the active arm, the movement of liquid may have been experienced as a positive stimulus and thus reinforced the responding of control planaria. There is some evidence dating back more than a century that planaria will actively swim against a current (Allen, 1915) which may indicate that moving water is preferred to still water. In partial support of this, it was observed throughout the project that when the water in the planarian housing was changed, planaria became more motile. However, whether this represents approach or avoidance behaviour is unknown.

In all cases where planaria became more likely to enter the active arm, be it for drug treated or control subjects, the proportion of active arm entries floated around 0.5. While one interpretation suggests that this change is evidence of learning among treatment subjects, an alternative explanation could be that planaria were exhibiting truly random behaviour. For this view, the baseline preferences, which hovered around 0.3 in most cases, could be the consequence of observing a small number of trials of the random behaviour. The bias that results from under sampling can be illustrated by a simple example such as flipping a coin. Using a simple unbiased coin flipping script in R, when we observed six coin flips per trials, four repeated trials produced the following outcomes: 5:1, 3:3, 6:0 and 4:2 (ratio of tails to heads). Meanwhile, observing 100,000 coin flips in a single trial resulted in 49.6% of the flips being tails and 50.4% being heads. Provided enough observations are made, the stochastic nature of the variable is revealed. That our apparent biases at baseline may simply be under sampling of a random variable is supported by the results of Abbott and Wong (2008) who found that when looking at a single baseline session containing ten trials, most planaria showed arm preferences in a Y-maze procedure. However, when combining thirty baseline trials across three days, most planaria showed no arm preference.

If Abbott and Wong (2008) are correct in claiming that planaria typically do not have an arm preference, we must then explain why the proportion of active arm entries observed in our experiments were consistent across two separate baseline days. If the behaviour was truly random, one would expect greater variability between baseline day one and baseline day two. However, across experiments for both treatment and control groups, the proportion of entries into the active arm was approximately 0.3 on two consecutive baseline days.

Although the planaria used by Abbott and Wong (2008) may not exhibit a directional preference, given many documented behavioural differences among planaria species (Cochet-Escartin et al., 2015; DeBold et al., 1965; Mueller & Levin, 2002; Samuel et al., 2021), it may be that there are also species differences in directional preference. Perhaps the species used throughout this project differed in their behaviour from the Dugesia Tigrina used by Abbott and Wong (2008). A targeted investigation would be required to determine if the species of planaria used here exhibit a directional bias.

Moving now to the behaviour of control subjects, it is difficult to know whether such dramatic shifts in behaviour should be expected. For the most part, this is because there are few similar studies available for comparison. The modern literature on planaria behaviour in general conveys stable behaviour of the control group in paradigms such as CPP (Hutchinson et al., 2015; Jordan et al., 2023). However, early work by Corning (1966) demonstrated relatively high variability of planarian behaviour in a T-maze paradigm. In fact, the control group in Corning

(1966) demonstrated a noticeable increase in active arm preference across the first ten trials, with the active arm preference remaining between 0.45 and 0.5 for the remaining 70 trials. In the case of Corning (1966), despite this increase for the control group, the treatment groups entered the active arm between 60-65% of the time. In line with our observations from Experiment 3, Corning (1966) saw a rapid spike in active arm entries for treatment subjects within the first ten trials.

Two recent projects which aimed to shape directional preferences also found high variability among the behaviour of control subjects. Read (2021) observed that the percentage of entries into the active arm varied from ~25% at baseline to ~50% at the end of conditioning for the control group. Another investigation (unpublished data, Canales Laboratory) observed similar variation in active arm entries when subjects were treated with cocaine alongside compounds known to prevent cocaine seeking in rodents. All groups experienced a large jump in active arm entries on the first day of conditioning, hovering around 50% and then declining towards baseline levels. This resembles the behaviour seen in the control groups within the current project and, to some extent, matches the decline in active arm entries seen in the treatment groups.

The intertrial interval is one factor that may affect the extent of learning among planaria. It has been suggested elsewhere that planaria learn mazes most effectively when a 30-minute intertrial interval is used (Warren, 1965, p. 100). Moreover, larger intertrial intervals have been reported to mitigate the effects of fatigue from repeated trials (Best & Rubinstein, 1962; Lee, 1963). But the optimal intertrial interval may differ largely between tasks. Some classical conditioning procedures have had success when using an intertrial interval of one minute. Crawford et al. (1966) found that spaced trials (at least one minute between) were more effective than massed trials (only 30 seconds between).

The experiments employed here varied in their intertrial intervals. Experiment two had a shorter intertrial interval of 15 minutes, while the remaining experiments had intertrial intervals of 60 minutes or more. This was due to a change in procedure. In Experiment 2, six planaria were moved into temporary petri dishes and completed all of their trials before the next group of six started their first trial. Whereas in later experiments, planaria were taken straight from their 12-well housing compartments, with each planaria completing their first trial before any planaria started their second trial. Although the intertrial interval may play a role in the rate and extent of learning, we did not observe any obvious difference in results when comparing Experiment 2 with subsequent experiments.

Another factor which may impact the rate of learning is the drug concentration used. Across the experiments reported here, 10 µM or 20 µM of the rewarding compound was administered. While these doses are similar to those used in most studies of learning and addiction-like behaviour in planaria (Amaning-Kwarteng et al., 2017; Hutchinson et al., 2015; Nayak et al., 2016; Raffa et al., 2006; Sacavage et al., 2008; Turel, 2022), there are a several papers which have employed drug concentrations as high as 80 µM with success (Raffa et al., 2005; Raffa & Desai, 2005; Rawls, Rodriguez, et al., 2006; Umeda et al., 2004).

As was observed in the dose-response analysis shown in Figure 5, there was no evidence that treatment with 10 µM or 20 µM of cocaine affected planaria motility. We specifically sought a concentration of cocaine that would not affect the movement of planaria during subsequent trials. But the lack of physical effects may also indicate that the drug is failing to have any psychoactive (and therefore rewarding) effects for the planaria. Although higher concentrations may reduce the speed with which planaria complete the Y-maze and increase the rate of non-responses, it may also increase the extent of learning. That said, some experiments have shown successful learning with drug concentrations as low as 1 µM (Hutchinson et al., 2015; Vouga et al., 2015). Given this, it would be beneficial to systematically manipulate the drug concentration used to reward planaria to identify an optimal dose which maximises learning in this species of planaria.

Given the instability of planarian behaviour observed here, it is difficult to recommend the Y-maze procedure as a viable conditioning paradigm for the field. If the phenomena of memory retention through regeneration is a reliable effect as is suggested by the literature (Corning, 1966; McConnell et al., 1959; Mueller & Levin, 2002; Rhodes & Vierick, 2024; Samuel et al., 2021; Shimojo et al., 2022; Shomrat & Levin, 2013), understanding whether this extends to complex forms of memory is a worthwhile pursuit. But to achieve this, a reliable method for effectively shaping planaria behaviour is needed. The Y-maze procedure may not be a reliable method. Instead, alternative operant conditioning procedures may be better suited to carry on this research project. There are several alternative methods for conditioning planarian.

Some date back to the early 20th century such as the Van Oye maze (e.g., Oye, 1920), while others have only appeared in the last decade. For example, Chicas-Mosier and Abramson (2015) established a method where the directed movement of planaria is reinforced with water in a crescent petri dish. Although independent replications of these methods are needed to demonstrate their viability, they may hold more promise for successfully shaping planaria behaviour.

One key insight from the work performed here is that planaria are highly variable in their behaviour. As was seen when assessing planaria motility during the dose-response procedure, there was large variability within each group. Some planaria covered the diameter of the dish just two or three times during the 15-minute recording interval, while many others travelled a distance 20 times greater than the dish diameter. Both of these extremes were observed across four of the five groups. Regarding the Y-maze, there was high variability in active arm entries over time even among the control group. When we observed an increase in active arm entries among methamphetamine treated subjects, this change was not stable and began to diminish rapidly towards the end of conditioning. Behavioural volatility may be a general characteristic of planaria.

There is evidence for between species differences in learning ability among planaria (Mueller & Levin, 2002; Samuel et al., 2021), and even within-species differences to slight changes in environmental conditions including light, vibrations, size of the recording dish and more (Rejo et al., 2023). If planarian behaviour exhibits high within-subject variability, the chance of finding a reliable operant conditioning procedure may be low.

Memory Retention through Regeneration

We shall now examine the more theoretically significant question addressed within this project: can a learned response persist through bisection and regeneration? The prevailing theory in the field, the synaptic trace theory of memory (Hebb, 1949; Poo et al., 2016), would suggest that a memory can only be retained if the synaptic connections which underpin it are retained. This theory would be challenged if a change in behaviour, such as a conditioned arm preference, is conserved in the tail regenerates of trained planaria. Our results showed that head regenerates of methamphetamine treated subjects maintained an active arm preference that was significantly higher than baseline. However, the tail regenerates failed to show retention of the active arm preference. A spontaneous change in the behaviour of controls was seen in regenerated head and tails, with both showing greater active arm entries compared to baseline.

The head regenerates of trained planaria should in theory contain the original brain cells present during the conditioning procedure. As synaptic trace theory predicts, the original dendritic spines that underpin the memory should not be affected by the bisection. The head regenerates could therefore act on previously acquired information. The predictions of synaptic trace theory are supported by the behaviour of head regenerates from methamphetamine exposed planaria in Experiment 4. Further support for the theory is provided by the tail regenerates. These regenerates did not differ significantly from baseline in their proportion of active arm entries. A proponent of the synaptic trace theory would argue that the necessary neural connections that underlay the memory were absent in the tail half and, therefore, the information could not possibly persist in the tail regenerates.

While the observed results for methamphetamine treated regenerates can be explained by the synaptic trace theory, the behaviour of control subjects is much harder to parse. The head and tail regenerates of control planaria in Experiment 4 exhibited a significantly higher proportion of active arm entries compared to baseline. That is, despite showing no evidence of learning during conditioning, the active arm entries increased significantly in both halves after bisection. Of note, the proportion of active arm entries in the head and tail regenerates of control subjects centred around 0.5, reflecting the earlier concern that planaria may not have a true directional preference. While it is tempting to claim that methamphetamine treated head regenerates demonstrated retention of a learned behaviour, the observed data cannot rule out that under sampling of a random behaviour at baseline is responsible for the pattern, rather than successful learning.

That the directional preference of planaria in a Y-maze may be random is supported by the findings of Akiyama et al. (2015). The authors found that a commonly observed behaviour in planaria, whereby they prefer to be on the wall of a dish rather than on the base of the dish, is due to spontaneous behaviours that increase the likelihood of ending up on the wall. The authors devised several experiments to show that, absent any alluring or noxious stimuli, planaria move straight ahead until they reach a wall. Moreover, they demonstrated that planaria perform a side-to-side movement of their head while swimming ("wigwag movements"), and that it is this spontaneous behaviour which affects their path of motion. The authors conclude that, what often appears to be intentional wall seeking behaviour may in fact be the result of two spontaneous behaviours. Despite this, planarian wall hugging behaviour is typically described in the literature as an intentional preference. But as Akiyama et al. (2015) suggested, planarian behaviour may be less intentional than initially presumed. If the wall hugging behaviour is the result of spontaneous movements rather than an intentional preference, it may have implications for planarian behaviour in the Y-maze. Specifically, it may imply that planaria do not have a true directional preference.

The handling technique used to transfer planaria may have contributed to the apparent behaviour changes observed across experiments. In the final three experiments, planaria were transferred into the Y-maze using a plastic transfer pipette. This method made it difficult to precisely control the starting position at the beginning of each trial. On occasion, a planarian would land close to or on a wall of the maze. While this did not guarantee that the planarian would enter a particular arm, if their default motion is to continue moving straight as suggested by Akiyama et al. (2015), it may have biased the outcome. This aligns with the experimenter's observations, as planaria seemed more likely to enter the arm corresponding to a wall it landed on or was closest to when entering the maze runway. Given the experimenter was right-handed, the discharge angle of planaria was typically biased towards the left hand wall. While most planaria landed on the floor of the maze runway, this may have increased the chance that planaria land on the left wall and enter the left arm. Fortunately, there was no evidence of a left-arm bias in planaria. In fact, when considering the baseline behaviour in Experiment 3 and 4, we found that subjects tended to enter the right arm more often at baseline. No systematic bias was seen in Experiment 5.

In summary, we may have effectively shaped the behaviour of treatment subjects in Experiment 3 and 4. But because the extent of behaviour change was limited and responses were unstable towards the end of conditioning, the experiments reported here can only be considered a weak test of the hypothesis. Additional experiments using more reliable training methods are required to explore whether operantly conditioned responses can persist through decapitation and regeneration. Although our results were inconclusive, they form part of a wider literature forcing us to question whether synaptic weights are the only game in town when it comes to storing memories.

We must contend with a whole host of research being amassed which has indicated that other memory storage mechanisms may be at play. These studies include those previously highlighted which demonstrate that simple associative memories can be maintained outside of the brain in planaria, and that these can be expressed once a new brain is regenerated (Corning, 1966; McConnell et al., 1959; Rhodes & Vierick, 2024; Samuel et al., 2021; Shimojo et al., 2022; Shomrat & Levin, 2013). But they also encompass work with other model organisms such as the nematode *C. elegans* and even single celled organisms such as *Paramecium caudatum* and *Physarum polycephalum* (which will be discussed shortly). Together, these studies compel us to turn our attention away from the crown jewel of memory research (synaptic plasticity) in search of other mechanisms that could facilitate memory storage. We shall now turn our attention to some alternative memory storage mechanisms that have been suggested, and discuss the applications and implications that may stem from a better understanding of these mechanisms.

Alternative Memory Storage Mechanisms and their Implications

Neuroscientists have focused their attention on neurons as the storehouse of memory for good reasons. There is compelling evidence that, by manipulating neurons, we can impact whether an animal succeeds or fails to learn a desired response. Moreover, these manipulations can even affect the lifespan of the memory. By introducing particular compounds or blocking the activation of certain proteins, researchers can manipulate whether a learned behaviour is rapidly forgotten or forms a long term memory (Bourtchuladze et al., 1994).

Early experiments allowed researchers to identify the biological pathways involved in memory. But recent innovative methods are allowing for greater levels of control over memory creation and expression by specifically manipulating neural cells. For example, J.-H. Han et al. (2009) selectively extinguished a fear memory by destroying the neurons active during fear acquisition. This showed that the neurons in the amygdala which were engaged when a new fear memory was formed can be tagged and selectively destroyed and that this inhibits the expression of the specific fear memory. An even greater level of precision was demonstrated by Liu et al. (2012). Using an optogenetic approach, which allows memory-associated neurons to be modified via light exposure, Liu et al. (2012) demonstrated that a fear response acquired in one context (by paring it with a shock) can be transferred to a novel context by simply activating the fear engram while the rodent is in that novel context. The rodents will then freeze in that context as if they had been previously shocked within it.

Clearly, neurons are important for memory. But can we confidently state that they are the only game in town? Before doing so, we must stop to ask: what other mechanisms might be capable of storing information from past experiences and thus shaping future behaviour?

Research in single celled organisms suggests that memories can be encoded without the use of neurons (see Gershman et al., 2021 for review). Single celled, by definition, implies there are no connections between cells. No connections between cells means no possibility of storing memories among synaptic weights. Yet, despite their lack of neurons, several studies have indicated that single celled organisms can demonstrate habituation and even classical conditioning. For example, Gelber (1952) found that paramecia, a unicellular ciliate, learned to congregate around a wire covered in nutritious bacteria that was placed in their culture dish. When a wire lacking any nourishing bacteria was later introduced during a test phase, the trained paramecia still gathered around it in larger numbers compared to untrained paramecia. More recently, Armus et al. (2006) have shown that paramecia can learn to associate the position of a cathode delivering an electric shock with a light. Further evidence of learning without the need for synaptic plasticity. Other unicellular organisms, such as the slime mold *Physarum polycephalum*, have also been shown to exhibit learning in the form of habituation to previously aversive compounds (Boisseau et al., 2016).

These studies on learning in single celled organisms do not identify what the underlying memory mechanisms are. They simply highlight, at a conceptual level, that a kind of learning can occur without neurons or synapses – proof by omission. Thankfully, there are a number of ongoing inquiries to understand how information, including experiences and other form of learning, may be encoded with the resources available to simple organisms.

RNA is best known for its role as the intermediary allowing our genes encoded in DNA to be translated into proteins. Although this may be its predominant role, it is just one of the many roles that RNA plays. Moore et al. (2021) have shown that RNAs can also play a role in communicating important survival information across generations and between organisms. *C.*

elegans will learn to avoid *P. aeruginosa*, a pathogenic bacteria, after exposure (Zhang et al., 2005). Incredibly, this learned response can be transferred to naive *C. elegans* by feeding them homogenised worms and can also can be passed to offspring for four generations.

Although *C. elegans* contains a nervous system, Moore et al. (2021) provided evidence that the information resulting in the aversion is stored in vesicles containing small RNAs. It is the tunnelling of these vesicles into gametes that is thought to transfer the experience transgenerationally. This indicates that, even if a brain is required to act on this information, the actual information can be stored outside of the central nervous system. This scenario may be far different than what we might expect when encountering the concept of memory transfer in the literature. For humans, it brings to mind the idea that our thoughts and experiences may literally be transferred to our offspring. An aversion a strain of bacteria is much simpler. Nevertheless, this work demonstrates that mechanisms beyond synaptic connections are capable of storing experiential information which can later be acted upon.

Bioelectric signalling provides another mechanism of information storage. Work on bioelectric signalling in planaria has shown that pattern memories, which hold information pertaining to how cells should arrange themselves to reach a target morphology, can be stored among the voltage gradients of regular somatic cells (Levin, 2023; Pezzulo et al., 2021). This means that the voltage gradients of cells can encode information regarding how the body should regenerate and what the end morphological goal is. By altering these voltage gradients, Beane et al. (2013) showed that planaria can be made to regenerate an additional head in place of their tail. A morphological pattern memory may be a long shot away from the kinds of experiential memories we are interested in. Moreover, whether a mechanism like bioelectric signalling is capable of storing experiential information attained by planaria, such as a texture or directional preference, is not yet known. Nevertheless, it provides a clear example of goal-directed information being stored outside of neural tissue and affecting future behaviour (regeneration).

Turning now to the final mechanism we will touch upon, epigenetic modifications have been shown to play a role in all forms of learning and memory, from habituation to fear conditioning (see Bronfman et al., 2016). The epigenetic mechanisms implicated cover all possible epigenetic markings. DNA methylation, histone modifications, and histone variations all modulate learning and memory performance (Bronfman et al., 2016; Perez & Lehner, 2019).

One epigenetic modification in particular, DNA methylation, was found to be necessary for learning to occur in rodents (J. Han et al., 2010; Heyward & Sweatt, 2015). The importance of DNA methylation was demonstrated by inhibiting methyltransferases which blocked memory formation (C. A. Miller & Sweatt, 2007). Histone acetylation, which opens up DNA and increases the rate of transcription, has also been associated with improved memory performance (Levenson et al., 2004).

Of particular importance for the current project, epigenetic regulation plays a crucial role in planaria during regeneration. For example, factors responsible for methylation and chromatin remodelling are critical for enabling successful regeneration (Bonuccelli et al., 2010; Hubert et al., 2013). In the context of learning and memory, it has already been suggested elsewhere that the pluripotent neoblast cells responsible for regeneration could possibly imprint previously learned information on the developing brain via epigenetic mechanisms (Blackiston et al., 2015). Though the details of memories maintained outside of the brain in planaria must be further explored, epigenetic regulation of neoblast cells may play a role.

The studies and mechanisms described above are not intended to suggest that neuronal connections are not the primary site of memory storage and expression. Instead, they serve to show that our current conception of memory storage is too narrow to account for findings available throughout the learning and memory literature (see Abraham et al., 2019 for a review which integrates traditional and non-synaptic mechanisms; see also Flores & Liester, 2024). Moreover, if non-synaptic storage mechanisms are exploited by simple organisms to store and act on information, we should consider whether these primitive mechanisms, whatever their structure may be, might also be at work in humans. That said, we have not yet addressed a question of central importance: why should we care if other memory storage mechanisms exist? It could be argued that we are already making progress in understanding and diagnosing conditions like Alzheimer's and other age related memory impairments. What benefits are to be gained by recognising alternative memory storage mechanisms?

If it was discovered that the body uses macromolecules such as RNAs or some other mechanism alongside neuronal storage, this would have important therapeutic and clinical implications. Primarily, the information stored outside of neurons could be exploited to reinstantiate information that has been disturbed after a traumatic event. People experience head trauma from a variety of activities and events. Contact sports, traffic accidents, and stroke are just a few common examples. Fortunately, some memories and abilities that at first appear lost will recover spontaneously over time (Kerr et al., 2011). But much of what is lost, be it motor abilities or past experiences, never returns.

Current efforts to improve rehabilitation after damage focus on task specific training and factors such as exercise which have been linked to improved patient outcomes (P. Han et al., 2017; Paris, 2007). But a greater understanding of non-synaptic storage mechanisms could improve recovery rates by facilitating the restoration of this knowledge. Our current view of memory storage suggests that we should be trying to maximise neurogenesis and structural reorganisation of neural tissue to improve recovery rates for patients with brain damage. But another route to recovery may lie in directing the movements of macromolecular storage components. This may also open up new avenues for peer assisted therapeutics. Just as microbiota can be transferred between individuals to enhance health outcomes (Mazzawi et al., 2018), molecular transfusions may similarly enable information to be moved between individuals to enhance cognitive outcomes. This is admittedly highly speculative, but the work of Moore et al. (2021) provides preliminary evidence that molecules such as RNAs can be transferred between organisms to confer the recipient with adaptive information that affects its behaviour.

Limitations

This project suffered from a number of limitations, some of which have been discussed already. A major constraint was that we had not carried out species level identification of the planaria used throughout this project. Given there were two phenotypes visible in the breeding colony, these may have represented two separate species. This limits the comparability of the results presented here with others in the literature and may even limit comparability between studies carried out within our lab. This is especially true given inter-species differences have been described for a number of behaviours and conditioning paradigms (Cochet-Escartin et al., 2015; DeBold et al., 1965; Mueller & Levin, 2002; Rejo et al., 2023; Samuel et al., 2021).

The concentration of drug administered during the Y-maze experiments could not be precisely controlled. This stemmed from two factors. First, when we attempted to ascertain the amount of liquid left in each arm after the plug had been inserted, there were slight variations each time. Second, when planaria were transferred using the plastic transfer pipette, there was always some additional planaria water being introduced along with the subject (despite a persistent effort to minimise this). While these two factors are not expected to affect the concentration by more than one or two micromolar for most trials, some trials may have experienced greater variation. Although a higher drug concentration was not expected to hinder learning, a particularly low concentration for a given trial may have done so.

Furthermore, because the drugs were dissolved in planaria water rather than being injected directly into each subject, the amount of drug absorbed by each subject was unknown. It is often said that planaria lack a circulatory system and uptake chemicals and nutrients in the water via epithelial absorption and diffusion (Felix et al., 2019; Lewallen & Burggren, 2020; Vu et al., 2015). We can assume this is how planaria took up the compounds we administered into the water. But uncertainty remains regarding whether the compounds would have reached the brain in the short period of time given for absorption.

This matter of drug uptake is further complicated as the location of drug action is different for different compounds. For example, Pagán et al. (2013) demonstrated that planaria require an intact brain to react to cocaine but not nicotine. In bisected tail fragments, exposure to nicotine but not cocaine produced seizure like movements. This may affect how rapidly the rewarding properties take effect. This difference could be modulated by the size of planaria, particularly in cases where receptors are found widely dispersed throughout the body. Because the planaria used throughout this report were not precisely weighed or measured, there may have been size-dependent differences in drug uptake which affected learning rates. To control for such effects, future experiments should consider ensuring subjects meet a specific size criterion.

Variability of the intertrial interval may have also affected learning for treatment subjects. For Experiments 3, 4 and 5, all subjects completed their first trial before any subjects started their second trial. Any given trial could take between just over three minutes up to eight minutes. This variability resulted in inconsistent intervals between trials and is the reason that only approximate intertrial intervals were provided. Moreover, the duration of drug exposure for planaria was imprecise due to running multiple subjects simultaneously. While we attempted to minimise variability in this regard, some planaria would have been exposed to the drug for longer than others. In some instances, this resulted in between one and two minutes of additional exposure time.

Another major limitation results from the small number of observations used to establish the baseline arm preference. This was already discussed extensively throughout the manuscript. Nevertheless, it must be reiterated. It is still questionable whether planaria show a directional preference in the Y-maze. By observing only six trials, we risked inferring a preference where no preference existed. Adding additional baseline trials would restrict the sample size (given the time-demanding nature of an operant conditioning procedure), but it would give a more reliable estimate of the initial directional preference. Having a robust baseline measure would improve the validity of interpreting later behaviour change as evidence of learning.

The large number of subjects excluded during Experiment 2 may have impacted the results. For the baseline to endpoint comparison, 9 control subjects were excluded compared to just 3 treatment subjects. Due to additional deaths during regeneration, the control group had just 18 subjects for the two week follow up test and 17 subjects for reinstatement. In comparison, the treatment group had 24 subjects at both follow up test points. The difference in the subject dropout rate may have contributed to the between-group differences detected at the test phase. Further analysis would be required to determine if the behaviour of those that were excluded was different from the remaining subjects.

A final limitation to the regeneration experiments reported here relates to the structure of tail fragments. While most of the planarian central nervous system is contained within the head in the form of a centralised brain, there are ventral nerve cords throughout the tail which are thought to contain neurons that form neural networks independent of the brain (Okamoto et al., 2005). The bisection should have successfully removed all of the brain tissue from the tail fragments, but would have left some of these posterior neural networks intact. A proponent of the synaptic trace theory could argue that the memories are still being stored synaptically in the tail halves of bisected planaria. For a clearer test of whether memories can be stored non-synaptically, which is to say outside of neural tissue, one would need to cut planaria in such a way that the final fragments lack any tissue from the nerve cords. It has been previously suggested that a planarian fragment around 1/279th the weight of the original worm can survive and regenerate (Morgan, 1898). With some others reporting that just 10,000 cells are required for complete cephalic regeneration (Montgomery & Coward, 1974). This may permit the use of smaller sections from the side of the body for the retention tests. This approach would test memory retention while ensuring synaptic storage mechanisms are ruled out.

Summary and Future Directions

Research carried out using popular model organisms such as rodents, birds and apes, enable straightforward inferences regarding how human memory operates. However, research using these animals suffers from many restrictions on the kinds of manipulations that can be performed. Other limitations arise due to the high cost of housing and maintaining these animals. Planaria present a unique opportunity to investigate the nature of memory, as they provide a means for investigating learning and memory phenomena with high-throughput, low cost and a wide scope for exploratory research. Given their regenerative abilities, planaria can be used to answer questions unavailable when working with typical model organisms, including whether memory can be retained outside of the brain.

The current project built on previous findings showing that simple associative memories can be retained in planaria after decapitation and regeneration of the brain. The experiments carried out here asked whether this phenomenon can be extended to more complex forms of memory, such as reward seeking behaviour. Although we found preliminary evidence that planaria could obtain a conditioned directional preference in a Y-maze, there was no evidence that this could persist in the tail regenerates. This failure does not definitively show that complex memories cannot be stored outside of the brain. Rather, it indicates that more robust procedures are required to improve learning rates. This will allow for a stronger test of the hypothesis.

Whether or not planaria are a viable organism for understanding addictive behaviours in humans directly depends on whether they can reliably learn complex tasks. Although a number of papers have explored planaria as a model for addiction (Amaning-Kwarteng et al., 2017; Francis, 2015; Mohammed Jawad et al., 2018; Samuel et al., 2021; Turel, 2022), simple texture preferences and light preferences are very far removed from the complex drug seeking behaviours found in humans and other mammals (Singer et al., 2018). If we want to use planaria as a model for understanding addiction related behaviours like habit formation, extinction, and tolerance, we must use models which have some resemblance to the behaviour we hope to inform in humans.

Humans must string together a number of behaviours and actions in the world to find and procure drugs of abuse. Therefore, to gather insight into the mechanisms governing these behaviours, we must at least capture some of the complexity of drug seeking behaviour in our animal models. If planaria can be reliably shown to acquire operantly conditioned responses, this should generate optimism for planaria as a viable model organism. On the other hand, if we fail to find a robust way to shape planaria behaviour, it is not clear that we will gain any insights of practical utility for addressing addiction and other behavioural issues in humans.

A number of suggested actions arise naturally from the lessons learnt during this project. First, for the continuation of planaria research in New Zealand, species level identification should be carried out to determine whether the genome of the planaria used here match those of other known species, or whether they are a novel species indigenous to New Zealand. Once the species used for this project has been identified, it will help position this work within the existing literature. Given the already described inter-species differences, it will help contextualise our failure to find strong evidence for operant conditioning. If we are working with a species native to New Zealand, it may be that they are simply poor learners. In contrast, if we are working with a species shared by other labs around the world, it may be a cause for skepticism around current claims of operant conditioning in the literature.

It remains to be shown whether the Y-maze is a viable procedure for studying learning and memory retention. A number of manipulations could be tested to optimise the procedure, allowing for a more robust test of whether complex memories can be stored outside of the brain. First, a range of doses could be used which test whether low or high concentrations are more effective for shaping behaviour, while allowing the researcher to observe whether high doses impact behaviour on subsequent trials (e.g. maze completion time). The following concentrations of both cocaine and methamphetamine could be tested: 1, 20, 50 and 150 μM. Once a concentration which maximises learning has been identified, manipulation of the exposure time should be carried out. The current experiment used a 3-minute absorption period. However, a longer duration may enhance learning. Absorption durations ranging from 5 to 10 minutes could be tested. A longer absorption period may constrain the sample size given the additional time required to complete all the trials. Nevertheless, if the behavioural change can be stabilised, this would be an acceptable trade-off.

As an alternative approach, one could search for another viable operant conditioning procedure. The Van Oye maze described in the literature review would be a useful starting point. The benefit of the Van Oye maze is that the task requires more movement and a larger sequence of behaviours than the Y-maze, which would make evidence of learning more obvious. In the van Oye Maze, at minimum a planaria must climb across the floor, up the wall, across the water surface, and down the fishing line to reach the food. The problems described earlier regarding baseline performance are also mitigated with the Van Oye setup. Despite being touted as one of the most successful operant conditioning paradigms (Nicolas et al., 2008), we could not find modern replications using the Van Oye method in the literature. Future experiments should attempt to replicate results reported in the 20th century (Corning & Riccio, 1970; Oye, 1920). Ideally, preregistration should be completed prior to experimentation which details what a successful trial looks like, the training protocol, exclusion criteria, and a power analysis to determine the required sample size to replicate effects reported previously.

Once a method for establishing effective operant conditioning has been reached, future experiments should consider whether it is viable to use a small fragment of trained planaria to test for memory retention. This will allow for a stronger test of the hypothesis that memories can be stored non-synaptically, as the neural tissue in the regenerating fragment can be minimised. Small fragments can be compared to tail fragments and or head fragments. If learning can persist in head and tail fragments but not the smaller fragments, then one may conclude that memory is stored outside of the centralised brain, but still maintained among synaptic connections in the ventral nerve cords.

Conclusion

This project used planaria to test whether complex forms of memories – those involved in reward seeking behaviour – can be stored outside the brain. Although there was some evidence that planaria will navigate to an arm where they were previously rewarded with a stimulant, this behaviour was not stable. Moreover, some of the control groups demonstrated a shift in behaviour similar to that of the treatment groups. Because of this, we could not make any strong claims about the learning ability of planaria or whether complex memories can be retained through decapitation and regeneration. Future research should focus on either improving the strength of memories in the Y-maze procedure or attempt to establish strong memories using other procedures previously reported in the literature.

Data Availability

This manuscript was developed using Quarto – a publishing software that allows for the creation of reproducible research papers. The datasets used to perform the analyses reported here, as well as the R scripts required to generate the statistics and figures, are available in the following GitHub Repository.

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